

Polyphase, Operator, and Jump-Measure Representations of the Fabius–Rvachev System

**A corpus-aware representation atlas with sparse-scale sinc products,
multidimensional section integrals, and new Thue–Morse formulas**

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Abstract

The recursive LaTeX corpus under `Analysis/FabiusFunction/docs` already contains a large fraction of the modern analytic theory of the Fabius function, its inverse, Rvachev’s compactly supported up-function, the Thue–Morse sequence, dyadic and Gaussian q -binomial arithmetic, sinc products, spectral determinants, Bell–Bernoulli moment algebra, and periodic Lambert– W endpoint asymptotics. The purpose of this report is therefore twofold: first, to organize the existing material as a single representation atlas; second, to add a complementary layer that was not located in the audited corpus.

The new organizing object is the sparse-scale Fourier product

$$\Psi_r(z) = \prod_{m \geq 0} \operatorname{sinc}_\pi\left(\frac{z}{2^m}\right), \quad r \in \mathbb{N}_{>0}.$$

Partitioning the scales of the ordinary Rvachev product modulo r gives the exact polyphase factorization

$$\widehat{\operatorname{up}}(z) = \prod_{a=0}^{r-1} \Psi_r(z/2^a).$$

If g_r is the probability density with Fourier image Ψ_r , then

$$\operatorname{up} = g_r * (2g_r(2 \cdot)) * \cdots * (2^{r-1}g_r(2^{r-1} \cdot)).$$

This one identity generates cosine, Gamma, canonical-zero, logarithmic-zeta, hyperbolic-product, cumulant, Bell-polynomial, spectral-zeta, and integral representations. The component zero multiplicity is $1 + \lfloor \nu_2(n)/r \rfloor$, and its spectral zeta is $\zeta(s)/(1 - 2^{-rs})$. Mellin analysis shows an exact roots-of-unity cancellation: the r -periodic pole lattice of a sparse component recombines so that only the ordinary dyadic harmonics survive.

The probability factorization yields a new family of finite-dimensional representations. With $w_r = (1, 2^{-1}, \dots, 2^{-(r-1)})$ and independent coordinates having density g_r , the Fabius function is both a sublevel-set volume and a parallel hyperplane section:

$$F(x) = \int_{w_r \cdot y \leq 2x-1} \prod_{a=0}^{r-1} g_r(y_a) \, dy = \frac{1}{\|w_r\|_2} \int_{w_r \cdot y = x-1} \prod_{a=0}^{r-1} g_r(y_a) \, d\sigma(y).$$

The equality of a moving volume and a differently placed section is the multidimensional form of the self-CDF identity $F(x) = \operatorname{up}(x-1)$.

A second new package extracts the Thue–Morse sequence from analytic data. Finite sparse box splines $g_{r,N}$ have knots $x_{r,N,k} = -A_{r,N} + \sum_j b_j(k)2^{-rj}$, and their highest classical derivative has jumps

$$\varepsilon_k = (-1)^{s_2(k)} = 2^{-r \binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}.$$

Equivalently, the complete signed block is the inverse Fourier transform of a deconvolution quotient:

$$\sum_{k < 2^N} \varepsilon_k \delta_{x_{r,N,k}} = 2^{-r \binom{N}{2}} D^N \mathcal{F}^{-1} \left[\frac{\Psi_r(z)}{\Psi_r(z/2^{rN})} \right].$$

Taking the positive and negative Jordan parts gives equally explicit formulas for the unsigned sequence. Pairing this measure with high powers gives exact Bell–Bernoulli formulas for all surviving Thue–Morse power sums and new limit representations for every moment of g_r and up.

Finally, convolution is recast as an infinite product of averaging operators,

$$\text{up} = \exp \left(\sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!(1-4^{-m})} D^{2m} \right) \delta_0,$$

with a rigorous polynomial interpretation and a strong-operator interpretation through uniform averaging. Commented Python code checks the polyphase products, Fourier deconvolution, exact power sums, and cumulant recombination. Throughout, “new” means new relative to the pinned repository snapshot; no claim of worldwide bibliographic priority is made without a separate exhaustive literature search.

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1 Scope, corpus audit, and novelty discipline

1.1 Pinned source boundary

The branch moved during the audit, so the mathematical source boundary is the exact commit printed on the title page. The recursive documentation tree was classified into five strata:

1. the primary theorem-backed exposition and mathematical glossary;
2. the living non-formalized frontier synthesis;
3. active thematic reports on Thue–Morse, spectra, arithmetic rays, exponent sequences, transforms, inverse sampling, multiresolution, and Fourier decay;
4. frozen historical lineages under `archive/`;
5. imported or corrected primary-paper sources under `papers/`.

Every TeX path was included in the recursive inventory. Superseded or nearly identical archive revisions were treated through their provenance lineages rather than counted as independent mathematical discoveries. The canonical current descendants were used as the source of record, while targeted full-corpus searches were made for every proposed new structure.

The most relevant exclusion sources are the primary Fabius–Rvachev synthesis [10], the unified Thue–Morse atlas [11], the finite-block calculus [12], the representation-frontier report [13], the multiresolution report [14], the inverse/sampling synthesis [15], the exponent-sequence synthesis [16], and the spectral-arithmetic synthesis [17]. The imported literature was also checked, notably Fabius’s original probabilistic example, Rvachev’s compact support theory, Arias de Reyna’s analytic and arithmetic papers, and the lacunary-product literature [2, 5, 3, 4, 9].

1.2 Status convention

The report uses four status tags.

Tag	Meaning
K	Explicit in the audited repository or classical source; retained to make the atlas usable.
D	A short transform, specialization, or recombination of known material; proved for completeness but not promoted as a substantive novelty.
N	Apparently absent from the pinned TeX corpus; a proof is supplied here.
C	Conjectural or proposed research direction; numerical evidence is identified separately.

1.3 What is deliberately not claimed as new

The following substantial structures already occur in the corpus and are therefore used only as baseline:

- the probability law of a geometrically weighted uniform series, finite atomic and spline approximants, the Fabius/up fold, and quantile transport;
- the sinc and triangular cosine products, Fourier inversion, critical Poisson cosine series, partition of unity, and exact integer-zero jets;
- exact dyadic values, Gaussian q -binomial formulae, Thue–Morse products, Prouhet cancellation, Mahler equations, Dirichlet–Mellin theory, and the large signed/unsigned formula atlas;
- sharp Fourier decay, log-periodic fluctuations, arithmetic rays, half-integer spectra, cyclotomic traces, spectral determinants, heat traces, and Barnes/Gamma dualities;
- Cauchy–Stieltjes resolvents, logarithmic finite-level resolvents, Jacobi and Stieltjes continued fractions, and Fredholm determinant models;

- Haar, Faber–Schauder, Walsh–Hadamard, beta-mixture, sinc-mixture, Fourier–Legendre, and inverse self-sampling representations;
- integer and fractional integration packages, inverse endpoint reversion, Bell-polynomial coefficient calculus, and periodic lower-Lambert asymptotics.

Novelty boundary in one sentence

The new contribution is not another isolated formula for the same sinc product. It is a modular-scale factorization that simultaneously produces a sparse Rvachev family, a Mellin alias-cancellation law, multidimensional volume–section identities, and analytic jump formulas for the signed and unsigned Thue–Morse sequences.

2 Normalization and structural dictionary

2.1 The probability law, the bump, and the bounded function

Let $(U_n)_{n \geq 0}$ be independent random variables uniformly distributed on $[-1, 1]$, and set

$$X = \sum_{n=0}^{\infty} 2^{-n-1} U_n. \quad (2.1)$$

The series converges absolutely almost surely, $X \in [-1, 1]$, and its density is Rvachev’s even bump

$$u(x) := \text{up}(x), \quad \int_{-1}^1 u(x) \, dx = 1, \quad u(0) = 1. \quad (2.2)$$

Let

$$Y = \frac{X + 1}{2} \in [0, 1]. \quad (2.3)$$

The bounded Fabius function is the distribution function of Y :

$$F(x) = \mathbb{P}(Y \leq x) = \mathbb{P}(X \leq 2x - 1), \quad 0 \leq x \leq 1. \quad (2.4)$$

The special self-similarity of the law gives the stronger pointwise fold

$$F(x) = u(x - 1), \quad F(1 - x) = u(x), \quad 0 \leq x \leq 1. \quad (2.5)$$

Thus the same function value is at once a probability of a half-line and a density value at a different point.

The inverse Fabius function is denoted

$$G = F^{-1} : (0, 1) \longrightarrow (0, 1). \quad (2.6)$$

Endpoint totalizations can be added when needed; no ambiguity below depends on them.

2.2 Fourier and sinc conventions

The Fourier transform is

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i z x} \, dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(z) e^{2\pi i z x} \, dz. \quad (2.7)$$

Write

$$\text{sinc}_{\pi}(z) = \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}_{\pi}(0) = 1, \quad \Phi(z) = \widehat{u}(z). \quad (2.8)$$

Then

$$\Phi(z) = \prod_{n=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{z}{2^n}\right), \quad \Phi(z) = \operatorname{sinc}_{\pi}(z)\Phi(z/2). \quad (2.9)$$

The moment-generating function is

$$M(s) = \mathbb{E}e^{sX} = \Phi\left(\frac{is}{2\pi}\right) = \prod_{n=0}^{\infty} \frac{\sinh(s/2^{n+1})}{s/2^{n+1}}. \quad (2.10)$$

2.3 Thue–Morse normalizations

For $k \geq 0$, write

$$\tau(k) = s_2(k) \bmod 2 \in \{0, 1\}, \quad \varepsilon_k = (-1)^{s_2(k)} = 1 - 2\tau(k). \quad (2.11)$$

The signed normalization is natural for products, differences, and atomic measures. The unsigned normalization is recovered by

$$\tau(k) = \frac{1 - \varepsilon_k}{2}. \quad (2.12)$$

3 A compact representation atlas

This section records the main representation families already available in the corpus, plus immediate transforms needed later. It is intentionally compact: the detailed novelty begins in [section 4](#). The tags refer to the convention of [section 1](#).

3.1 Probability, infinite-cube, and distribution representations

From (2.1), for every bounded Borel function h ,

$$\int_{-1}^1 h(x)u(x) \, dx = \mathbb{E} h\left(\sum_{n \geq 0} 2^{-n-1}U_n\right), \quad (3.1)$$

$$F(x) = \int_{[-1,1]^{\mathbb{N}}} \mathbf{1}_{\{\sum 2^{-n-1}v_n \leq 2x-1\}} \prod_{n \geq 0} \frac{dv_n}{2}. \quad (3.2)$$

These are [K](#). They immediately imply the quantile-transport formula

$$\int_0^1 h(G(y)) \, dy = \int_0^1 h(x) \, dF(x) = \mathbb{E} h(Y), \quad (3.3)$$

again [K](#).

3.2 Differential and refinement representations

The bump satisfies

$$u'(x) = 2(u(2x+1) - u(2x-1)), \quad (3.4)$$

and F satisfies

$$F'(x) = 2F(2x) \quad (0 \leq x \leq 1/2), \quad F(1-x) = 1 - F(x). \quad (3.5)$$

Together with (2.5), these give two useful inverse-derivative forms:

$$G'(y) = \frac{1}{F'(G(y))} = \frac{1}{u'(G(y)-1)} = \frac{1}{2u(2G(y)-1)}. \quad (3.6)$$

All are [K](#).

3.3 Fourier inversion and CDF inversion

Rapid decay of Φ gives

$$u(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z x} dz = 2 \int_0^\infty \Phi(z) \cos(2\pi z x) dz, \quad (3.7)$$

$$F(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z(x-1)} dz, \quad 0 \leq x \leq 1. \quad (3.8)$$

Applying the Gil–Pelaez inversion formula to X gives the companion CDF form

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(z) \frac{\sin(2\pi z(2x-1))}{z} dz. \quad (3.9)$$

The equality of (3.8) and (3.9) is a Fourier form of the self-CDF identity. Status: K/D.

3.4 Laplace, Bromwich, and Fourier images of the bounded function

The Laplace–Stieltjes transform of F is

$$\int_0^1 e^{sx} dF(x) = \mathbb{E}e^{sY} = e^{s/2} M(s/2). \quad (3.10)$$

Integration by parts therefore yields, for $s \neq 0$,

$$\int_0^1 e^{sx} F(x) dx = \frac{e^s - e^{s/2} M(s/2)}{s}, \quad (3.11)$$

with limiting value $1/2$ at $s = 0$. Setting $s = -2\pi i \xi$ gives the exact Fourier image

$$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}, \quad (3.12)$$

again continued by $1/2$ at the origin. Conversely, for $c > 0$ and $0 < x < 1$,

$$F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{e^{s(x-1/2)}}{s} M(s/2) ds. \quad (3.13)$$

These formulas are K or immediate one-line transforms of known identities.

3.5 Product, canonical-zero, and Gamma representations

The audited corpus contains the following equivalent products:

$$\Phi(z) = \prod_{n=0}^{\infty} \operatorname{sinc}_{\pi}(z/2^n), \quad (3.14)$$

$$= \prod_{m=1}^{\infty} \cos(\pi z/2^m)^m, \quad (3.15)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}, \quad (3.16)$$

$$= \prod_{n=0}^{\infty} \frac{1}{\Gamma(1+z/2^n)\Gamma(1-z/2^n)}. \quad (3.17)$$

Thus the nonzero integer n is a zero of order $1 + \nu_2(n)$. The local logarithmic series is

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1-4^{-m})} z^{2m}, \quad |z| < 1, \quad (3.18)$$

and the meromorphic logarithmic derivative is

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^2 - z^2}. \quad (3.19)$$

Status: **K**.

3.6 Moments, cumulants, and Bell–Bernoulli series

Writing

$$\log M(s) = \sum_{m \geq 1} \kappa_{2m} \frac{s^{2m}}{(2m)!}, \quad (3.20)$$

comparison with (2.10) gives

$$\kappa_{2m} = \frac{B_{2m}}{2m(1-4^{-m})}, \quad \kappa_{2m+1} = 0. \quad (3.21)$$

Hence

$$\mu_n := \int x^n u(x) \, dx = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (3.22)$$

where $\mathcal{B}_n$ is the complete exponential Bell polynomial. In particular

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \kappa_4 = -\frac{2}{225}. \quad (3.23)$$

Status: **K**.

3.7 Finite splines, Poisson series, and multiresolution forms

The corpus supplies finite uniform-convolution splines whose coefficients are Thue–Morse signs, critical Poisson cosine series on $[-1, 1]$, exact partition of unity, Fourier–Legendre expansions, and rational Haar, Walsh, and Faber–Schauder systems. Representative formulas are

$$u(x) = \frac{1}{2} + \sum_{k \geq 0} \Phi(k + \frac{1}{2}) \cos((2k+1)\pi x), \quad (3.24)$$

$$\sum_{j \in \mathbb{Z}} u(x-j) = 1, \quad (3.25)$$

with the convergence and endpoint conventions established in the primary source. The multiresolution report expresses the same function as limits of exact rational step mixtures, Haar details, Walsh finite products, Bernstein beta mixtures, and inverse-quantile sinc mixtures [14]. Status: **K**.

3.8 Resolvent and determinant forms

Let

$$R(z) = \int_{-1}^1 \frac{u(x)}{z-x} \, dx, \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (3.26)$$

Then

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}} \quad (3.27)$$

for $|z| > 1$, and the representation-frontier report derives a logarithmic fixed point, a Thue–Morse derivative tree, finite logarithmic resolvents, and the Jacobi continued fraction of R [13]. Likewise (3.16) is the Fredholm determinant

$$\Phi(z) = \det(I - z^2 T), \quad (3.28)$$

where T is diagonal with eigenvalue n^{-2} repeated $1 + \nu_2(n)$ times. Status: K.

Transition

The formulas above are a map, not the claimed contribution. The remainder of the report introduces a decomposition by congruence classes of the scale index. It simultaneously refines the Fourier product, the probability law, the zero divisor, the Mellin pole lattice, the finite splines, and the Thue–Morse coefficient measure.

4 Polyphase factorization by scale classes

4.1 Sparse-scale products and laws

Fix an integer $r \geq 1$ and put

$$q_r = 2^{-r}, \quad \Psi_r(z) = \prod_{m=0}^{\infty} \text{sinc}_{\pi}(q_r^m z). \quad (4.1)$$

Let Z_r be the random series

$$Z_r = \sum_{m=0}^{\infty} \frac{q_r^m}{2} U_m, \quad (4.2)$$

where the U_m are independent uniform variables on $[-1, 1]$. Its support radius is

$$A_r = \frac{1}{2(1 - q_r)}, \quad (4.3)$$

and its density is denoted g_r . Then

$$\widehat{g}_r(z) = \Psi_r(z), \quad \text{supp } g_r = [-A_r, A_r]. \quad (4.4)$$

The generic geometrically weighted uniform law is part of the repository baseline. What is new here is its use as a polyphase factor of the ordinary Rvachev law.

Theorem 4.1 (Polyphase product and convolution factorization). *For every integer $r \geq 1$,*

$$\Phi(z) = \prod_{a=0}^{r-1} \Psi_r(z/2^a). \quad (4.5)$$

Equivalently, if

$$g_{r,a}(x) = 2^a g_r(2^a x) \quad (4.6)$$

is the density of $2^{-a} Z_r$, then

$$u = g_{r,0} * g_{r,1} * \cdots * g_{r,r-1}. \quad (4.7)$$

In random-variable form,

$$X \stackrel{d}{=} \sum_{a=0}^{r-1} 2^{-a} Z_r^{(a)}, \quad (4.8)$$

where the $Z_r^{(a)}$ are independent copies of Z_r .

Proof. Every nonnegative integer has a unique decomposition $n = a + rm$ with $0 \leq a < r$ and $m \geq 0$. Therefore

$$\prod_{n \geq 0} \operatorname{sinc}_\pi(z/2^n) = \prod_{a=0}^{r-1} \prod_{m \geq 0} \operatorname{sinc}_\pi\left(\frac{z}{2^{a+rm}}\right) = \prod_{a=0}^{r-1} \Psi_r(z/2^a).$$

The products converge locally uniformly, so regrouping is legitimate. The Fourier transform of $g_{r,a}$ is $\Psi_r(z/2^a)$; multiplication of the transforms gives (4.7). The random-series identity is the same partition of the summands of (2.1). $\square$

Remark 4.2. The support radii add exactly:

$$\sum_{a=0}^{r-1} 2^{-a} A_r = \frac{1}{2(1-2^{-r})} \sum_{a=0}^{r-1} 2^{-a} = 1. \quad (4.9)$$

Thus no slack is introduced by the factorization.

Example 4.3 (The even–odd scale split). For $r = 2$,

$$\Phi(z) = \Psi_2(z) \Psi_2(z/2), \quad u = g_2 * (2g_2(2 \cdot)), \quad (4.10)$$

where g_2 is supported on $[-2/3, 2/3]$. One factor contains scales $1, 1/4, 1/16, \dots$ in the full widths; the other contains the intervening scales after one dyadic contraction.

4.2 A sparse Rvachev differential equation

The product (4.1) satisfies

$$\Psi_r(z) = \operatorname{sinc}_\pi(z) \Psi_r(q_r z). \quad (4.11)$$

The random fixed point is

$$Z_r \stackrel{d}{=} \frac{U}{2} + q_r Z'_r, \quad (4.12)$$

with independent $U \sim \operatorname{Unif}[-1, 1]$ and $Z'_r \stackrel{d}{=} Z_r$.

Proposition 4.4 (Sparse-scale Rvachev equation). *The density g_r obeys*

$$g_r(x) = \int_{2^r x - 2^{r-1}}^{2^r x + 2^{r-1}} g_r(t) dt, \quad (4.13)$$

$$g'_r(x) = 2^r (g_r(2^r x + 2^{r-1}) - g_r(2^r x - 2^{r-1})). \quad (4.14)$$

For $r = 1$ this is the ordinary Rvachev equation.

Proof. Condition on $U/2$ in (4.12). The density of $q_r Z'_r$ is $q_r^{-1} g_r(\cdot/q_r)$, hence

$$g_r(x) = \int_{-1/2}^{1/2} q_r^{-1} g_r\left(\frac{x-y}{q_r}\right) dy.$$

The substitution $t = (x - y)/q_r$ gives (4.13); differentiation gives (4.14). $\square$

4.3 Cosine, Gamma, canonical-zero, and logarithmic forms

The sparse product has a complete set of analogues of the ordinary Rvachev representations.

Theorem 4.5 (Equivalent products for the sparse component). *For every $r \geq 1$,*

$$\Psi_r(z) = \prod_{m=0}^{\infty} \operatorname{sinc}_{\pi}(z/2^{rm}), \quad (4.15)$$

$$= \prod_{n=1}^{\infty} \cos(\pi z/2^n)^{1+\lfloor (n-1)/r \rfloor}, \quad (4.16)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\lfloor \nu_2(n)/r \rfloor}, \quad (4.17)$$

$$= \prod_{m=0}^{\infty} \frac{1}{\Gamma(1+z/2^{rm})\Gamma(1-z/2^{rm})}. \quad (4.18)$$

For $|z| < 1$,

$$\log \Psi_r(z) = - \sum_{j=1}^{\infty} \frac{\zeta(2j)}{j(1-2^{-2rj})} z^{2j}. \quad (4.19)$$

Away from its integer zeros,

$$\frac{\Psi'_r(z)}{\Psi_r(z)} = -2z \sum_{n=1}^{\infty} \frac{1 + \lfloor \nu_2(n)/r \rfloor}{n^2 - z^2}. \quad (4.20)$$

Proof. The sinc and Gamma forms follow from $\operatorname{sinc}_{\pi} w = (\Gamma(1+w)\Gamma(1-w))^{-1}$. Expanding each sinc by $\operatorname{sinc}_{\pi} w = \prod_{j \geq 1} \cos(\pi w/2^j)$, a fixed cosine at scale 2^{-n} occurs once for each $m \geq 0$ with $rm < n$, giving $1 + \lfloor (n-1)/r \rfloor$ copies. Expanding instead by Euler's sine product, a zero at a nonzero integer n receives one copy for each m with $rm \leq \nu_2(n)$, hence multiplicity $1 + \lfloor \nu_2(n)/r \rfloor$. The logarithmic series follows from $\log \operatorname{sinc}_{\pi} z = - \sum_{j \geq 1} \zeta(2j) z^{2j}/j$ and a geometric sum. Logarithmic differentiation of the canonical product gives (4.20). $\square$

Corollary 4.6 (Exact splitting of the integer zero divisor). *Let $v = \nu_2(n)$. The zero order $1 + v$ of Φ at $n \neq 0$ decomposes as*

$$1 + v = \sum_{a=0}^{r-1} \mathbf{1}_{a \leq v} \left(1 + \left\lfloor \frac{v-a}{r} \right\rfloor\right). \quad (4.21)$$

The a th summand is the order contributed by $\Psi_r(z/2^a)$.

Proof. The right side counts pairs (a, m) satisfying $0 \leq a < r$, $m \geq 0$, and $a + rm \leq v$. Uniqueness of Euclidean division identifies these pairs with the integers $0, 1, \dots, v$. $\square$

4.4 Spectral zeta and determinant splitting

Let

$$m_r(n) = 1 + \left\lfloor \frac{\nu_2(n)}{r} \right\rfloor. \quad (4.22)$$

The zero divisor of Ψ_r has spectral zeta

$$\mathcal{Z}_r(s) = \sum_{n \geq 1} \frac{m_r(n)}{n^s}. \quad (4.23)$$

Proposition 4.7 (Sparse spectral zeta). *For $\Re s > 1$,*

$$\mathcal{Z}_r(s) = \frac{\zeta(s)}{1 - 2^{-rs}}. \quad (4.24)$$

Moreover the scaled factors satisfy the exact recombination

$$\sum_{a=0}^{r-1} 2^{-as} \mathcal{Z}_r(s) = \frac{\zeta(s)}{1 - 2^{-s}}, \quad (4.25)$$

which is the spectral zeta of Φ .

Proof. Each factor $\text{sinc}_\pi(z/2^{rm})$ contributes one zero at every nonzero multiple of 2^{rm} . Therefore

$$\mathcal{Z}_r(s) = \sum_{m \geq 0} \sum_{k \geq 1} (2^{rm}k)^{-s} = \zeta(s) \sum_{m \geq 0} 2^{-rms}.$$

Scaling the zeros by 2^a multiplies their zeta contribution by 2^{-as} . The finite geometric sum cancels $1 - 2^{-rs}$. $\square$

In particular,

$$\Psi_r(z) = \det(I - z^2 T_r), \quad (4.26)$$

where T_r has eigenvalue n^{-2} repeated $m_r(n)$ times. The ordinary Rvachev determinant is the direct-sum product of the r scaled component operators.

4.5 Hyperbolic products, cumulants, and Bell polynomials

The component moment-generating function is

$$M_r(s) = \mathbb{E} e^{sZ_r} = \Psi_r\left(\frac{is}{2\pi}\right) = \prod_{m=0}^{\infty} \frac{\sinh(s/2^{rm+1})}{s/2^{rm+1}}. \quad (4.27)$$

Theorem 4.8 (Sparse cumulants and exact recombination). *The component cumulants are*

$$\kappa_{2j}^{(r)} = \frac{B_{2j}}{2j(1 - 2^{-2rj})}, \quad \kappa_{2j+1}^{(r)} = 0. \quad (4.28)$$

The $2j$ th cumulant of the polyphase sum (4.8) is

$$\sum_{a=0}^{r-1} 2^{-2aj} \kappa_{2j}^{(r)} = \frac{B_{2j}}{2j(1 - 2^{-2j})} = \kappa_{2j}. \quad (4.29)$$

Consequently

$$\mu_n^{(r)} := \mathbb{E} Z_r^n = \mathcal{B}_n(\kappa_1^{(r)}, \dots, \kappa_n^{(r)}). \quad (4.30)$$

Proof. Use

$$\log \frac{\sinh x}{x} = \sum_{j \geq 1} \frac{2^{2j-1} B_{2j}}{j(2j)!} x^{2j}$$

in (4.27) and sum the resulting geometric progression. The scaling law for cumulants and the finite geometric sum give (4.29). The Bell-polynomial relation between moments and cumulants is standard. $\square$

The first component values are

$$\text{Var } Z_r = \frac{1}{12(1 - 2^{-2r})}, \quad \kappa_4^{(r)} = -\frac{1}{120(1 - 2^{-4r})}. \quad (4.31)$$

For $r = 2$, for example,

$$\kappa_2^{(2)} = \frac{4}{45}, \quad \kappa_4^{(2)} = -\frac{32}{3825}. \quad (4.32)$$

The weighted sum of the two dilated cumulants recovers $1/9$ and $-2/225$.

Contribution I: one object, many representations

The sparse product Ψ_r is simultaneously a Fourier image, a Gamma product, a cosine product with staircase exponents, a canonical product with quotient valuation multiplicities, a Fredholm determinant, a hyperbolic moment product, and the transform of a sparse Rvachev density. Its r dyadic dilates multiply or convolve to the classical object. This modular-scale package was not found as such in the pinned corpus.

5 Mellin alias cancellation and log-periodic phases

The polyphase identity does more than regroup convergent products. It explains how the denser dyadic pole lattice is synthesized from a sparse one.

5.1 A general harmonic-sum lemma

Let $h : (0, \infty) \rightarrow \mathbb{C}$ be such that its Mellin transform

$$H(s) = \int_0^\infty h(t)t^{s-1} dt \quad (5.1)$$

exists in a vertical strip with $\Re s < 0$. Define

$$L_r(t) = \sum_{m=0}^{\infty} h(2^{-rm}t). \quad (5.2)$$

When necessary, polynomial or logarithmic principal parts are subtracted before applying this statement, exactly as in standard Mellin analysis of harmonic sums [6].

Theorem 5.1 (Polyphase Mellin cancellation). *In every common strip of absolute convergence,*

$$\mathcal{M}[L_r](s) = \frac{H(s)}{1 - 2^{rs}}, \quad (5.3)$$

$$L_1(t) = \sum_{a=0}^{r-1} L_r(2^{-a}t), \quad (5.4)$$

$$\mathcal{M}[L_1](s) = \frac{H(s)}{1 - 2^s}. \quad (5.5)$$

At a pole

$$s_k = \frac{2\pi i k}{r \log 2} \quad (5.6)$$

created by $1 - 2^{rs}$, the scale-sum multiplier is

$$\sum_{a=0}^{r-1} 2^{as_k} = \sum_{a=0}^{r-1} e^{2\pi i k a/r} = \begin{cases} r, & r \mid k, \\ 0, & r \nmid k. \end{cases} \quad (5.7)$$

Thus recombination cancels every sparse harmonic except every r th one.

Proof. For one term, $t = 2^{rm}u$ gives

$$\int_0^\infty h(2^{-rm}t)t^{s-1} dt = 2^{rms}H(s).$$

Summation yields (5.3). Euclidean division of the scale index gives (5.4); the Mellin transform of $L_r(2^{-a}t)$ is $2^{as}\mathcal{M}[L_r](s)$. The finite geometric numerator cancels $1 - 2^{rs}$ and gives (5.5). Evaluating the numerator on the pole lattice gives the roots-of-unity sum. $\square$

5.2 Application to logarithmic Fourier and Laplace products

Take, after the usual local regularization,

$$h(t) = \log \operatorname{sinc}_\pi(t) \quad \text{or} \quad h(t) = \log \frac{\sinh(t/2)}{t/2}. \quad (5.8)$$

Then L_r is respectively $\log \Psi_r$ or $\log M_r$. The component has a logarithmic-scale pole lattice with fundamental period $r \log 2$; the product of its r dyadic dilates retains only the harmonics compatible with period $\log 2$. This is the Mellin-side analogue of (4.21) and (4.29).

Corollary 5.2 (Periodic coefficient filter). *Suppose the regularized asymptotic expansion of a sparse component contains a Fourier series in the logarithmic phase*

$$P_r(\theta) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k \theta / r}. \quad (5.9)$$

Then the corresponding coefficient after polyphase recombination is obtained by the exact projector

$$P_1(\theta) = \sum_{a=0}^{r-1} P_r(\theta - a) = r \sum_{\ell \in \mathbb{Z}} c_{r\ell} e^{2\pi i \ell \theta}. \quad (5.10)$$

Remark 5.3. The existing Fourier-decay campaign identifies nontrivial orbit-dependent and periodic behavior of the full product. The theorem above does not replace that analysis. It factors its periodic data into a sparse phase and an exact cyclic projection, suggesting a new way to compute or certify the fluctuation by working at the longer period $r \log 2$ and then filtering.

Contribution II: Mellin polyphase projector

Partitioning the product by scale residues induces a literal roots-of-unity filter on Mellin residues. Zero multiplicities, cumulants, and log-periodic asymptotic harmonics are three manifestations of the same finite cyclic projection.

6 Multidimensional volume and section representations

The polyphase random decomposition turns the one-dimensional self-CDF identity into a relation between two different geometric operations on the same product density: integration over a half-space and integration over a parallel hyperplane.

6.1 Radon notation and the product density

Let

$$w_r = (1, 2^{-1}, \dots, 2^{-(r-1)}) \in \mathbb{R}^r, \quad p_r(y) = \prod_{a=0}^{r-1} g_r(y_a). \quad (6.1)$$

For an integrable function p on $\mathbb{R}^r$, define its cumulative and ordinary Radon transforms in direction $w \neq 0$ by

$$\mathcal{C}_w p(t) = \int_{w \cdot y \leq t} p(y) \, dy, \quad (6.2)$$

$$\mathcal{R}_w p(t) = \frac{1}{\|w\|_2} \int_{w \cdot y = t} p(y) \, d\sigma_{r-1}(y). \quad (6.3)$$

The coarea formula gives $(\mathcal{C}_w p)' = \mathcal{R}_w p$ at every continuity point. For the smooth compactly supported density p_r , this holds everywhere.

Theorem 6.1 (Polyphase Radon representation). *For every $r \geq 1$ and every $t \in \mathbb{R}$,*

$$u(t) = \mathcal{R}_{w_r} p_r(t), \quad \mathbb{P}(X \leq t) = \mathcal{C}_{w_r} p_r(t). \quad (6.4)$$

Consequently, for $0 \leq x \leq 1$,

$$F(x) = \mathcal{C}_{w_r} p_r(2x - 1) = \mathcal{R}_{w_r} p_r(x - 1). \quad (6.5)$$

a relation between a half-space volume at level $2x - 1$ and a parallel section at level $x - 1$.

Proof. By [theorem 4.1](#), the linear form $w_r \cdot (Z_r^{(0)}, \dots, Z_r^{(r-1)})$ has the same law as X . Its distribution function is therefore the cumulative Radon transform in [\(6.4\)](#). The coarea formula identifies its density with the ordinary Radon transform. Substitute the defining relation $F(x) = \mathbb{P}(X \leq 2x - 1)$ and the self-CDF identity $F(x) = u(x - 1)$. $\square$

Remark 6.2. At $r = 1$, [\(6.5\)](#) is exactly the one-dimensional identity

$$\int_{-\infty}^{2x-1} u(t) \, dt = u(x - 1).$$

For $r > 1$, the same identity becomes genuinely multidimensional: the left side is an r -dimensional sublevel volume, while the right side is a weighted $(r - 1)$ -dimensional slice.

6.2 Coordinate integral forms

Let

$$H_r(t) = \int_{-\infty}^t g_r(v) \, dv \quad (6.6)$$

be the CDF of the sparse component. Eliminating the last coordinate in the half-space and in the hyperplane gives two explicit $(r - 1)$ -fold integrals.

Corollary 6.3 (Volume–section integral pair). *For $r \geq 2$ and $0 \leq x \leq 1$,*

$$F(x) = \int_{\mathbb{R}^{r-1}} \left(\prod_{a=0}^{r-2} g_r(y_a) \right) H_r \left(2^{r-1} \left(2x - 1 - \sum_{a=0}^{r-2} 2^{-a} y_a \right) \right) dy, \quad (6.7)$$

$$F(x) = 2^{r-1} \int_{\mathbb{R}^{r-1}} \left(\prod_{a=0}^{r-2} g_r(y_a) \right) g_r \left(2^{r-1} \left(x - 1 - \sum_{a=0}^{r-2} 2^{-a} y_a \right) \right) dy. \quad (6.8)$$

Both integrals are effectively over a compact box because g_r is compactly supported.

Proof. In the volume formula, condition on $y_0, \dots, y_{r-2}$ and solve the inequality for y_{r-1} . In the section formula, solve $w_r \cdot y = x - 1$ for the last coordinate. The Jacobian is 2^{r-1} . $\square$

For the first nontrivial split $r = 2$, writing H_2 for the CDF of g_2 ,

$$F(x) = \int_{\mathbb{R}} g_2(y) H_2(4x - 2 - 2y) \, dy, \quad (6.9)$$

$$= 2 \int_{\mathbb{R}} g_2(y) g_2(2x - 2 - 2y) \, dy. \quad (6.10)$$

Thus an integral involving the CDF of g_2 at one affine scale equals an integral involving its density at another.

6.3 A geometric representation of the inverse Fabius function

The preceding theorem also gives an implicit representation of $G = F^{-1}$ that uses only one fixed product density and moving parallel hyperplanes.

Corollary 6.4 (Two-level geometry of the quantile). *For $0 < y < 1$, the unique $x = G(y)$ is characterized by*

$$\boxed{\mathcal{C}_{w_r} p_r(2x - 1) = \mathcal{R}_{w_r} p_r(x - 1) = y.} \quad (6.11)$$

Its derivative is the reciprocal of a third section:

$$\boxed{G'(y) = \frac{1}{2\mathcal{R}_{w_r} p_r(2G(y) - 1).}} \quad (6.12)$$

Equivalently, the same product density has section mass y at level $G(y) - 1$, while the section at level $2G(y) - 1$ controls the quantile elasticity.

Proof. Equation (6.11) is (6.5) with $x = G(y)$. Since $F'(x) = 2u(2x - 1)$, the inverse-function theorem and (6.4) give (6.12). $\square$

The quantile-transport transform becomes, for every integrable test function h ,

$$\int_0^1 h(G(y)) \, dy = \int_{\mathbb{R}^r} h\left(\frac{1 + w_r \cdot z}{2}\right) p_r(z) \, dz. \quad (6.13)$$

In particular,

$$\int_0^1 e^{sG(y)} \, dy = e^{s/2} \prod_{a=0}^{r-1} M_r(s/2^{a+1}), \quad (6.14)$$

which reduces, by cumulant recombination, to $e^{s/2} M(s/2)$.

6.4 Polyphase transforms of the bounded Fabius function

Substitution of (4.5) into the transform formulas of section 3 gives product–integral representations involving only the sparse component:

$$F(x) = 2 \int_0^\infty \left(\prod_{a=0}^{r-1} \Psi_r(z/2^a) \right) \cos(2\pi z(x - 1)) \, dz, \quad (6.15)$$

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \left(\prod_{a=0}^{r-1} \Psi_r(z/2^a) \right) \frac{\sin(2\pi z(2x - 1))}{z} \, dz, \quad (6.16)$$

$$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = \frac{e^{-\pi i \xi} \prod_{a=0}^{r-1} \Psi_r(\xi/2^{a+1}) - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (6.17)$$

These are D, but they make the new factorization directly usable for evaluating F rather than only u .

6.5 Nested polyphase factorizations

The scale-class decomposition is coherent under further subdivision.

Theorem 6.5 (Divisibility-lattice factorization). *For positive integers r and s ,*

$$\Psi_r(z) = \prod_{a=0}^{s-1} \Psi_{rs}(z/2^{ra}), \quad (6.18)$$

$$g_r = \bigast_{a=0}^{s-1} (2^{ra} g_{rs}(2^{ra} \cdot)). \quad (6.19)$$

The factorization is associative: subdividing first into classes modulo r and then each class modulo s gives exactly the same factors as a direct split modulo rs .

Proof. Write the sparse scale index as $m = a + s\ell$ and regroup the locally uniformly convergent product. Fourier inversion gives the convolution formula. $\square$

On spectral zeta functions the same coherence is the finite geometric identity

$$\frac{\zeta(z)}{1 - 2^{-rz}} = \sum_{a=0}^{s-1} 2^{-raz} \frac{\zeta(z)}{1 - 2^{-rsz}}. \quad (6.20)$$

Thus the family $(\Psi_r)_{r \geq 1}$ forms a compatible factor system over the divisibility poset.

Contribution III: a volume–section identity

Every modular-scale factorization of the Fourier product yields an r -dimensional product density whose cumulative Radon transform at $2x - 1$ and ordinary Radon transform at $x - 1$ are the same Fabius value. The inverse function is the unique position where these two moving geometric quantities coincide at a prescribed level.

7 Finite sparse splines and Thue–Morse jump measures

Fix $r \geq 1$, put $q = 2^{-r}$, and let $N \geq 1$. Define the finite random sum

$$Z_{r,N} = \sum_{j=0}^{N-1} \frac{q^j}{2} U_j, \quad (7.1)$$

with density $g_{r,N}$. Its support radius and subset-sum knots are

$$A_{r,N} = \frac{1 - q^N}{2(1 - q)}, \quad (7.2)$$

$$\lambda_{r,N}(k) = \sum_{j=0}^{N-1} b_j(k) q^j, \quad (7.3)$$

$$x_{r,N,k} = -A_{r,N} + \lambda_{r,N}(k), \quad 0 \leq k < 2^N. \quad (7.4)$$

Here $b_j(k)$ is the j th binary digit of k .

7.1 Truncated-power and Fourier-quotient formulas

Theorem 7.1 (Finite sparse box spline). *The density $g_{r,N}$ is the degree- $(N-1)$ compact spline*

$$g_{r,N}(x) = \frac{q^{-\binom{N}{2}}}{(N-1)!} \sum_{k=0}^{2^N-1} \varepsilon_k (x - x_{r,N,k})_+^{N-1}. \quad (7.5)$$

Its Fourier transform has the equivalent finite-product and deconvolution forms

$$\widehat{g_{r,N}}(z) = \prod_{j=0}^{N-1} \text{sinc}_\pi(q^j z) = \frac{\Psi_r(z)}{\Psi_r(q^N z)}. \quad (7.6)$$

Moreover,

$$g_r = g_{r,N} * (q^{-N} g_r(\cdot/q^N)), \quad (7.7)$$

so the omitted tail is exactly a q^N -dilate of the full sparse law.

Proof. The density of the j th summand is $q^{-j} \mathbf{1}_{[-q^j/2, q^j/2]}$. The standard inclusion–exclusion formula for a convolution of nonuniform boxes gives (7.5); the product of all widths is $q^{\binom{N}{2}}$ and the left endpoints produce the knots (7.4). Multiplying the individual Fourier images gives the first part of (7.6), while splitting the infinite product at N gives the quotient. The random-series tail is $q^N Z'_r$ with $Z'_r \stackrel{d}{=} Z_r$, proving (7.7). $\square$

7.2 The derivative-jump formula

The N th distributional derivative of a normalized truncated power is a Dirac mass. Applying this to (7.5) exposes the entire Thue–Morse block as the singular derivative of one positive spline.

Theorem 7.2 (Thue–Morse jump measure). *In the space of compactly supported distributions,*

$$D^N g_{r,N} = q^{-\binom{N}{2}} \sum_{k=0}^{2^N-1} \varepsilon_k \delta_{x_{r,N,k}}. \quad (7.8)$$

Consequently the piecewise constant derivative $D^{N-1} g_{r,N}$ has jump

$$\varepsilon_k = q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}. \quad (7.9)$$

The unsigned bit is therefore

$$\tau(k) = \frac{1}{2} \left(1 - q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}} \right). \quad (7.10)$$

These formulas hold for every N large enough to contain the binary expansion of k .

Proof. Distributionally, $D^N (x-a)_+^{N-1} = (N-1)! \delta_a$. This gives (7.8). The mass of $D^N g_{r,N}$ at a knot is exactly the classical jump of $D^{N-1} g_{r,N}$ there, proving (7.9); (7.10) follows from $\tau = (1 - \varepsilon)/2$. $\square$

Corollary 7.3 (Smooth integral extraction). *Let φ_k be any C^N function satisfying $\varphi_k(x_{r,N,\ell}) = \delta_{k\ell}$. Then*

$$\varepsilon_k = q^{\binom{N}{2}} \langle D^N g_{r,N}, \varphi_k \rangle \quad (7.11)$$

$$= (-1)^N q^{\binom{N}{2}} \int_{\mathbb{R}} g_{r,N}(x) \varphi_k^{(N)}(x) dx. \quad (7.12)$$

One may take φ_k to be the Lagrange cardinal polynomial on the knot set. Thus each Thue–Morse sign has an exact integral representation against a positive spline.

7.3 Reading the sequence from left to right

Let $\text{rev}_N(j)$ denote reversal of the N binary digits of j . Since $q \leq 1/2$, each weight q^j exceeds the sum of all later finite weights, except for a strict finite remainder at $q = 1/2$. The spatial order of the knots is therefore lexicographic in $(b_0, b_1, \dots, b_{N-1})$.

Proposition 7.4 (Bit reversal and spatial order). *If the knots are listed increasingly as $x_{r,N}^\uparrow(0) < \dots < x_{r,N}^\uparrow(2^N - 1)$, then*

$$x_{r,N}^\uparrow(j) = x_{r,N,\text{rev}_N(j)}. \quad (7.13)$$

Because binary weight is invariant under reversal,

$$q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N}^\uparrow(j)} = \varepsilon_{\text{rev}_N(j)} = \varepsilon_j. \quad (7.14)$$

Hence scanning the jumps from the left endpoint to the right endpoint spells the ordinary length- 2^N signed Thue–Morse word in its usual order; replacing $+1, -1$ by $0, 1$ gives the unsigned word.

Proof. The first differing digit in the order $b_0, b_1, \dots$ determines the larger subset sum because $q^j > \sum_{\ell=j+1}^{N-1} q^\ell$ for $q < 1/2$, and the same inequality remains strict for the finite sum when $q = 1/2$. Lexicographic order in low-to-high binary digits is ordinary integer order after bit reversal. Finally, $s_2(\text{rev}_N(j)) = s_2(j)$. $\square$

7.4 Atomic convolution, Fourier, and Laplace products

Define the signed and unsigned knot combs

$$\Theta_{r,N} = \sum_{k < 2^N} \varepsilon_k \delta_{x_{r,N,k}}, \quad (7.15)$$

$$\Lambda_{r,N} = \sum_{k < 2^N} \delta_{x_{r,N,k}}. \quad (7.16)$$

They factor as convolutions of two-point measures:

$$\Theta_{r,N} = \bigast_{j=0}^{N-1} (\delta_{-q^j/2} - \delta_{q^j/2}), \quad (7.17)$$

$$\Lambda_{r,N} = \bigast_{j=0}^{N-1} (\delta_{-q^j/2} + \delta_{q^j/2}). \quad (7.18)$$

Their Fourier transforms are the finite lacunary products

$$\widehat{\Theta}_{r,N}(z) = (2i)^N \prod_{j=0}^{N-1} \sin(\pi q^j z) \quad (7.19)$$

$$= (2\pi i z)^N q^{\binom{N}{2}} \widehat{g_{r,N}}(z), \quad (7.20)$$

$$\widehat{\Lambda}_{r,N}(z) = 2^N \prod_{j=0}^{N-1} \cos(\pi q^j z). \quad (7.21)$$

Equivalently,

$$\Theta_{r,N} = q^{\binom{N}{2}} D^N \mathcal{F}^{-1} \left[\frac{\Psi_r(z)}{\Psi_r(q^N z)} \right]. \quad (7.22)$$

This is the announced Fourier deconvolution representation of the signed sequence.

The bilateral Laplace transforms are similarly

$$\int e^{tx} d\Theta_{r,N}(x) = (-2)^N \prod_{j=0}^{N-1} \sinh(tq^j/2), \quad (7.23)$$

$$\int e^{tx} d\Lambda_{r,N}(x) = 2^N \prod_{j=0}^{N-1} \cosh(tq^j/2). \quad (7.24)$$

More generally, the Hamming-weight enumerator is the master product

$$\sum_{k < 2^N} y^{s_2(k)} e^{tx_{r,N,k}} = e^{-tA_{r,N}} \prod_{j=0}^{N-1} (1 + ye^{tq^j}). \quad (7.25)$$

The choices $y = -1$ and $y = 1$ recover (7.23) and (7.24).

7.5 Unsigned evil and odious measures

Let

$$E_{r,N} = \sum_{\tau(k)=0} \delta_{x_{r,N,k}}, \quad (7.26)$$

$$O_{r,N} = \sum_{\tau(k)=1} \delta_{x_{r,N,k}}. \quad (7.27)$$

Then the Jordan decomposition is exact:

$$E_{r,N} = \frac{\Lambda_{r,N} + \Theta_{r,N}}{2}, \quad O_{r,N} = \frac{\Lambda_{r,N} - \Theta_{r,N}}{2}. \quad (7.28)$$

Accordingly,

$$\widehat{E}_{r,N}(z) = 2^{N-1} \left(\prod_{j < N} \cos(\pi q^j z) + i^N \prod_{j < N} \sin(\pi q^j z) \right), \quad (7.29)$$

$$\widehat{O}_{r,N}(z) = 2^{N-1} \left(\prod_{j < N} \cos(\pi q^j z) - i^N \prod_{j < N} \sin(\pi q^j z) \right). \quad (7.30)$$

These are product representations for the unsigned Thue–Morse partition on the sparse knot set. Individual membership is given by the jump formula (7.10); aggregate transforms are given by (7.29)–(7.30).

Contribution IV: Thue–Morse as analytic singular data

The signed and unsigned Thue–Morse sequences can be recovered pointwise from jumps of the highest classical derivative of a positive compact spline, measure-theoretically from its next distributional derivative, integrally by testing that derivative against cardinal functions, and spectrally from a quotient of two sparse infinite sinc products. In spatial order no external binary labels are needed: the jumps already occur in Thue–Morse order.

8 Bell–Bernoulli power sums on sparse knots

The jump measure immediately converts the moments of a positive box spline into high-degree signed power sums. This extends ordinary Prouhet cancellation past its first nonzero degree and gives every surviving sum in closed Bell form.

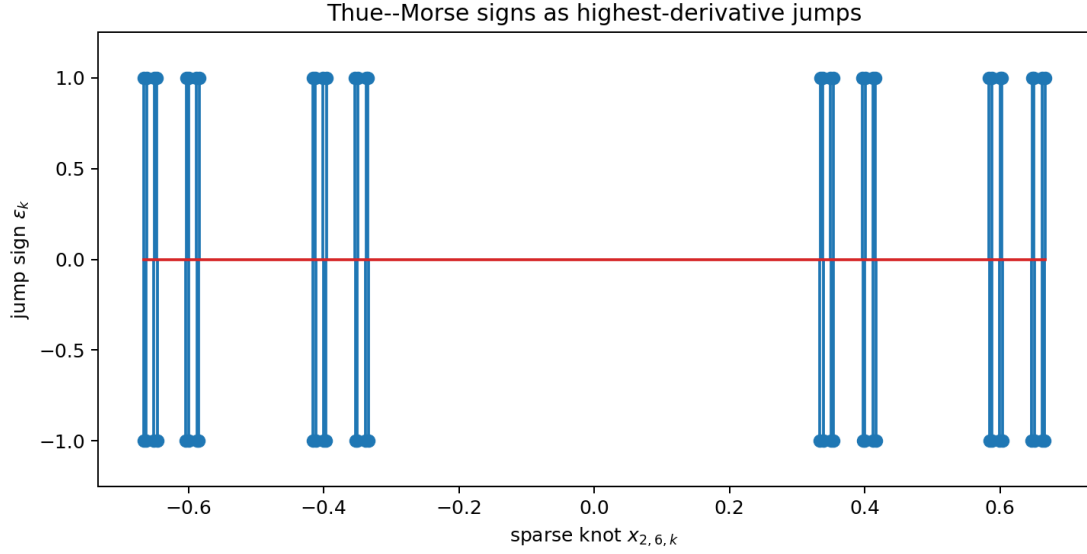


Figure 1: The distribution $q^{\binom{N}{2}} D^N g_{r,N}$ for a finite sparse spline. Reading its signed atoms from left to right gives the ordinary Thue–Morse word because the spatial bit-reversal leaves binary parity unchanged. The plotted example is generated by the accompanying script.

8.1 Prouhet cancellation and all surviving powers

Let

$$S_{r,N}(d) = \sum_{k=0}^{2^N-1} \varepsilon_k x_{r,N,k}^d. \quad (8.1)$$

Theorem 8.1 (Moment–power-sum duality). *For $0 \leq d < N$,*

$$S_{r,N}(d) = 0. \quad (8.2)$$

For every $m \geq 0$,

$$S_{r,N}(N+m) = (-1)^N q^{\binom{N}{2}} \frac{(N+m)!}{m!} \mu_m^{(r,N)}, \quad (8.3)$$

where

$$\mu_m^{(r,N)} = \int_{\mathbb{R}} x^m g_{r,N}(x) \, dx. \quad (8.4)$$

Equivalently,

$$\mu_m^{(r,N)} = (-1)^N q^{-\binom{N}{2}} \frac{m!}{(N+m)!} \sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+m}. \quad (8.5)$$

Odd m give zero by symmetry.

Proof. Pair (7.8) with the polynomial x^d . If $d < N$, its N th derivative vanishes. If $d = N + m$, distributional integration by parts gives

$$q^{-\binom{N}{2}} S_{r,N}(N+m) = (-1)^N \frac{(N+m)!}{m!} \int x^m g_{r,N}(x) \, dx.$$

Rearrange. □

The first two nontrivial cases are

$$\sum_{k < 2^N} \varepsilon_k x_{r,N,k}^N = (-1)^N N! q^{\binom{N}{2}}, \quad (8.6)$$

$$\sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+2} = (-1)^N q^{\binom{N}{2}} \frac{(N+2)!}{24} \frac{1 - q^{2N}}{1 - q^2}. \quad (8.7)$$

Thus the full hierarchy refines the usual statement that the first surviving Prouhet power occurs in degree N .

8.2 Finite cumulants and complete Bell form

The moment-generating function of $Z_{r,N}$ is

$$M_{r,N}(s) = \prod_{j=0}^{N-1} \frac{\sinh(sq^j/2)}{sq^j/2}. \quad (8.8)$$

Its cumulants are finite geometric truncations of the sparse cumulants.

Proposition 8.2 (Finite sparse cumulants). *For $\ell \geq 1$,*

$$\kappa_{2\ell}^{(r,N)} = \frac{B_{2\ell}}{2\ell} \frac{1 - q^{2\ell N}}{1 - q^{2\ell}}, \quad \kappa_{2\ell+1}^{(r,N)} = 0. \quad (8.9)$$

Hence

$$\mu_m^{(r,N)} = \mathcal{B}_m(0, \kappa_2^{(r,N)}, 0, \kappa_4^{(r,N)}, \dots). \quad (8.10)$$

Combining with (8.3) gives an explicit Bell–Bernoulli formula for every signed power sum $S_{r,N}(N+m)$.

In particular, for $\ell \geq 0$,

$$S_{r,N}(N+2\ell) = (-1)^N q^{\binom{N}{2}} \frac{(N+2\ell)!}{(2\ell)!} \mathcal{B}_{2\ell}(0, \kappa_2^{(r,N)}, 0, \dots, \kappa_{2\ell}^{(r,N)}). \quad (8.11)$$

This is a finite product–series–Bell representation in which every parameter is rational for $q = 2^{-r}$.

8.3 Unsigned power sums

The total knot comb $\Lambda_{r,N}$ is the unnormalized law of the centered Bernoulli sum

$$W_{r,N} = \sum_{j=0}^{N-1} q^j (B_j - \tfrac{1}{2}), \quad \mathbb{P}(B_j = 0) = \mathbb{P}(B_j = 1) = \tfrac{1}{2}. \quad (8.12)$$

Its even cumulants are

$$\lambda_{2\ell}^{(r,N)} = \frac{(2^{2\ell} - 1)B_{2\ell}}{2\ell} \frac{1 - q^{2\ell N}}{1 - q^{2\ell}}, \quad (8.13)$$

with odd cumulants zero. Therefore

$$T_{r,N}(d) := \sum_{k < 2^N} x_{r,N,k}^d = 2^N \mathcal{B}_d(0, \lambda_2^{(r,N)}, 0, \lambda_4^{(r,N)}, \dots). \quad (8.14)$$

If

$$T_{r,N}^{\text{evil}}(d) = \sum_{\tau(k)=0} x_{r,N,k}^d, \quad (8.15)$$

$$T_{r,N}^{\text{odious}}(d) = \sum_{\tau(k)=1} x_{r,N,k}^d, \quad (8.16)$$

then

$$T_{r,N}^{\text{evil}}(d) = \frac{T_{r,N}(d) + S_{r,N}(d)}{2}, \quad (8.17)$$

$$T_{r,N}^{\text{odious}}(d) = \frac{T_{r,N}(d) - S_{r,N}(d)}{2}. \quad (8.18)$$

For $d < N$ the two values coincide; for $d = N + m$ their exact difference is (8.3). Thus the unsigned partition inherits a complete Bell–Bernoulli moment calculus, not only the usual equality up to degree $N - 1$.

8.4 Infinite moment limits

Since $Z_{r,N} \rightarrow Z_r$ almost surely and uniformly in the underlying cube, all moments converge. Therefore (8.5) gives

$$\mu_m^{(r)} = \lim_{N \rightarrow \infty} (-1)^N q^{-\binom{N}{2}} \frac{m!}{(N+m)!} \sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+m}. \quad (8.19)$$

At $r = 1$, these are the moments of Rvachev’s up-function:

$$\int_{-1}^1 x^m u(x) \, dx = \lim_{N \rightarrow \infty} (-1)^N 2^{\binom{N}{2}} \frac{m!}{(N+m)!} \sum_{k < 2^N} \varepsilon_k x_{1,N,k}^{N+m}. \quad (8.20)$$

The enormous factor $q^{-\binom{N}{2}}$ exactly renormalizes the Prouhet cancellation.

Moments of the bounded Fabius law and powers of the inverse follow from $Y = (X + 1)/2$:

$$\int_0^1 G(y)^m \, dy = \mathbb{E}Y^m = 2^{-m} \sum_{j=0}^m \binom{m}{j} \mu_j, \quad (8.21)$$

$$\int_0^1 x^p F(x) \, dx = \frac{1 - \mathbb{E}Y^{p+1}}{p+1}, \quad \Re p > -1. \quad (8.22)$$

Substitution of (8.20) therefore yields explicit Thue–Morse limit representations for both integrals.

8.5 A Gaussian q -binomial side identity

The same binary digits carry a second geometric statistic $\sigma(k) = \sum_j j b_j(k)$. Its joint Hamming-weight enumerator is

$$\sum_{k < 2^N} y^{\sigma(k)} q^{\sigma(k)} = \prod_{j=0}^{N-1} (1 + yq^j) = \sum_{h=0}^N y^h q^{\binom{h}{2}} \begin{bmatrix} N \\ h \end{bmatrix}_q. \quad (8.23)$$

This familiar Gaussian-binomial identity is not the same as the knot transform (7.25): one weights a selected digit by q^j , the other places it at the geometric coordinate q^j . Their coexistence is useful in symbolic calculations because the first organizes selected scale indices, while the second organizes the actual spline knots.

Contribution V: all Prouhet survivors

The finite sparse spline does more than explain cancellation below degree N . Its moments give every surviving signed power sum in degree $N + m$, and finite uniform cumulants turn the answer into a closed Bell polynomial in Bernoulli numbers and geometric factors. Adding the cosh-product moments of the unsigned knot comb gives separate exact formulas for the evil and odious classes.

9 Infinite products of averaging operators

The Fourier product can be read literally as a product of physical-space averaging operators. This provides a representation which is simultaneously probabilistic, convolutional, and an infinite-order differential calculus.

9.1 Uniform averaging and strong product limits

For $a > 0$, define

$$(\mathcal{U}_a f)(x) = \frac{1}{2a} \int_{-a}^a f(x-t) dt. \quad (9.1)$$

Its Fourier multiplier and formal differential symbol are

$$\widehat{\mathcal{U}_a f}(z) = \text{sinc}_\pi(2az) \widehat{f}(z), \quad \mathcal{U}_a = \frac{\sinh(aD)}{aD}. \quad (9.2)$$

The second equality is exact on polynomials and on every class on which the translation exponential may be integrated termwise.

Put

$$\mathcal{G}_{r,N} = \prod_{j=0}^{N-1} \mathcal{U}_{q^j/2}, \quad \mathcal{G}_r = \prod_{j=0}^{\infty} \mathcal{U}_{q^j/2}. \quad (9.3)$$

The factors commute.

Theorem 9.1 (Averaging-operator representation). *For every bounded uniformly continuous f ,*

$$(\mathcal{G}_{r,N} f)(x) = (g_{r,N} * f)(x) = \mathbb{E} f(x - Z_{r,N}), \quad (9.4)$$

$$(\mathcal{G}_r f)(x) = (g_r * f)(x) = \mathbb{E} f(x - Z_r), \quad (9.5)$$

and $\mathcal{G}_{r,N} f \rightarrow \mathcal{G}_r f$ uniformly. The same convergence is strong on $L^p(\mathbb{R})$ for $1 \leq p < \infty$. Distributionally,

$$g_{r,N} = \mathcal{G}_{r,N} \delta_0, \quad g_r = \mathcal{G}_r \delta_0. \quad (9.6)$$

For $r = 1$ this gives

$$u = \left(\prod_{j=0}^{\infty} \mathcal{U}_{2^{-j-1}} \right) \delta_0. \quad (9.7)$$

Proof. Each operator convolves with the density of a centered uniform random variable of half-width $q^j/2$. Finite products therefore give the law of $Z_{r,N}$. The tail satisfies $|Z_r - Z_{r,N}| \leq A_r q^N$ pointwise on the underlying product space, which proves uniform convergence for uniformly continuous f . Strong L^p convergence follows from continuity of translations in L^p and dominated averaging. Applying the convolution operators to δ_0 gives the densities. $\square$

9.2 Bernoulli logarithm of the operator product

The classical expansion

$$\log \frac{\sinh z}{z} = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m} \quad (9.8)$$

shows that the logarithm of the sparse averaging product is controlled by the same Bernoulli cumulants as the random law.

Define

$$c_{r,m} = \frac{B_{2m}}{2m(2m)!(1 - q^{2m})}. \quad (9.9)$$

Theorem 9.2 (Infinite-order differential representation). *As a formal differential operator, and exactly on every polynomial,*

$$\mathcal{G}_r = \exp\left(\sum_{m=1}^{\infty} c_{r,m} D^{2m}\right), \quad (9.10)$$

$$\mathcal{G}_{r,N} = \exp\left(\sum_{m=1}^{\infty} c_{r,m} (1 - q^{2mN}) D^{2m}\right). \quad (9.11)$$

If f has Fourier support in a compact subset of $(-1, 1)$, then (9.10) is also an exact Fourier-multiplier identity. For the classical up-function,

$$u = \exp\left(\frac{D^2}{18} - \frac{D^4}{2700} + \frac{D^6}{178605} - \frac{D^8}{9639000} + \dots\right) \delta_0, \quad (9.12)$$

where the equality is understood through the exact averaging product and, termwise, through its formal or band-limited expansion.

Proof. From (9.2) and (9.8),

$$\log \mathcal{U}_{q^{j/2}} = \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} q^{2mj} D^{2m}.$$

Summing over j gives (9.10); truncating the geometric sum gives (9.11). On a polynomial only finitely many powers of D survive. On Fourier support $|z| < 1$, substitution $D \mapsto 2\pi iz$ gives precisely the convergent series (4.19). The displayed coefficients in (9.12) are $c_{1,m}$. $\square$

Using Euler’s formula for $\zeta(2m)$, the same operator can be written

$$\mathcal{G}_r = \exp\left(\sum_{m=1}^{\infty} \frac{(-1)^{m+1} \zeta(2m)}{m(1 - q^{2m})(2\pi)^{2m}} D^{2m}\right). \quad (9.13)$$

Thus the Bernoulli, zeta, cumulant, and Fourier-logarithm representations are literally the same coefficients in four coordinate systems.

9.3 Exact tail operators and series–product combinations

Let $\mathcal{T}_{r,N}$ be convolution by the scaled tail density $q^{-N} g_r(\cdot/q^N)$. Then

$$\mathcal{G}_r = \mathcal{T}_{r,N} \mathcal{G}_{r,N}, \quad \widehat{\mathcal{T}_{r,N} f}(z) = \Psi_r(q^N z) \widehat{f}(z). \quad (9.14)$$

On polynomials and in the low-frequency calculus,

$$\mathcal{T}_{r,N} = \exp\left(\sum_{m=1}^{\infty} c_{r,m} q^{2mN} D^{2m}\right). \quad (9.15)$$

Consequently, for every polynomial p ,

$$\int p(x - y) g_r(y) \, dy = \exp\left(\sum_{m \geq 1} c_{r,m} D^{2m}\right) p(x), \quad (9.16)$$

and

$$\int p(x - y) q^{-N} g_r(y/q^N) \, dy = \exp\left(\sum_{m \geq 1} c_{r,m} q^{2mN} D^{2m}\right) p(x). \quad (9.17)$$

Expanding the last exponential gives the exact finite identity

$$\mathbb{E}p(x - q^N Z_r) = \sum_{m=0}^{\deg p} \frac{(-q^N)^m \mu_m^{(r)}}{m!} p^{(m)}(x), \quad (9.18)$$

whose odd terms vanish. This is a rigorous series–integral representation, not an asymptotic expansion, whenever the test function is a polynomial.

For analytic or smooth functions the same coefficients organize the familiar small-tail expansion. The first two even orders are

$$\mathcal{T}_{r,N} = I + c_{r,1} q^{2N} D^2 + \left(c_{r,2} + \frac{c_{r,1}^2}{2} \right) q^{4N} D^4 + \mathcal{O}_{\text{formal}}(q^{6N} D^6). \quad (9.19)$$

The coefficient of $q^{2mN} D^{2m}$ is equivalently $\mu_{2m}^{(r)}/(2m)!$, by the Bell relation between cumulants and moments.

9.4 Reciprocal and deconvolution operators

Since $\widehat{g_{r,N}} = \Psi_r/\Psi_r(q^N \cdot)$, the finite spline is a formal deconvolution of the smooth limit:

$$g_{r,N} = \exp \left(- \sum_{m \geq 1} c_{r,m} q^{2mN} D^{2m} \right) g_r. \quad (9.20)$$

This identity is exact as a germ of Fourier multipliers at the origin and gives all algebraic orders in q^{2N} . Its global use requires the zero-aware factorization (7.6), because the reciprocal of $\Psi_r(q^N z)$ has poles which are cancelled by zeros of $\Psi_r(z)$. That distinction is important: the product quotient is globally exact, while the exponential reciprocal is a local or asymptotic coordinate chart.

The first logarithmic coefficients, generated exactly by the accompanying script, are shown in table 1.

$2m$	$c_{1,m}$	$c_{2,m}$	$c_{3,m}$
2	1/18	2/45	8/189
4	−1/2700	−4/11475	−64/184275
6	1/178605	64/11609325	4096/743175405
8	−1/9639000	−32/309652875	−8192/79272340875
10	1/478533825	1024/490497170625	1048576/502269581253825

Table 1: Coefficients of D^{2m} in $\log \mathcal{G}_r$. The complete exact table through order 16 is included as `operator_coefficients.csv`.

Contribution VI: an operator logarithm

The infinite sinc product is the Fourier symbol of an infinite product of uniform averaging operators. Its logarithm is an even differential operator whose coefficients are simultaneously sparse cumulants, Bernoulli numbers, and zeta values. On polynomials the exponential representation is exact; on band-limited functions it is an exact multiplier; globally the averaging product supplies the canonical interpretation.

10 Twisted Poisson and zero-slope representations

The critical half-integer cosine series is one member of a continuous family obtained by twisting Poisson summation. This family also yields a representation of F through the derivatives of Φ at its integer zeros.

10.1 A one-parameter Zak family

For real α , define

$$Z_u(x, \alpha) = \sum_{n \in \mathbb{Z}} u(x + n) e^{-2\pi i n \alpha}. \quad (10.1)$$

Rapid decay of Φ and compact support of u justify Poisson summation absolutely in the forms used below.

Theorem 10.1 (Twisted spectral representation of F). *For $0 \leq x \leq 1$,*

$$Z_u(x, \alpha) = \sum_{k \in \mathbb{Z}} \Phi(k + \alpha) e^{2\pi i (k + \alpha)x} \quad (10.2)$$

and also

$$Z_u(x, \alpha) = 1 + (e^{2\pi i \alpha} - 1)F(x). \quad (10.3)$$

Hence, whenever $\alpha \notin \mathbb{Z}$,

$$F(x) = \frac{\sum_{k \in \mathbb{Z}} \Phi(k + \alpha) e^{2\pi i (k + \alpha)x} - 1}{e^{2\pi i \alpha} - 1}. \quad (10.4)$$

Using polyphase factors, each coefficient may be replaced by $\prod_{a=0}^{r-1} \Psi_r((k + \alpha)/2^a)$.

Proof. The modulated Poisson formula gives (10.2). On the interval $0 < x < 1$, only the translates $n = 0$ and $n = -1$ meet the support of u , so

$$Z_u(x, \alpha) = u(x) + e^{2\pi i \alpha} u(x - 1) = 1 - F(x) + e^{2\pi i \alpha} F(x).$$

Continuity supplies the endpoint values. □

At $\alpha = 1/2$, (10.4) becomes the real cosine series

$$F(x) = \frac{1}{2} - \sum_{k=0}^{\infty} \Phi(k + \frac{1}{2}) \cos((2k + 1)\pi x), \quad (10.5)$$

which is the shifted form of (3.24). Rational values of α give cyclotomic spectral expansions; irrational values give a continuous family of quasiperiodic representations.

10.2 Differentiating through the untwisted limit

Both numerator and denominator of (10.4) vanish at $\alpha = 0$. Differentiation gives a new zero-jet representation.

Corollary 10.2 (Integer-zero slope series). *For $0 \leq x \leq 1$,*

$$F'(x) = x + \frac{1}{2\pi i} \sum_{k \in \mathbb{Z} \setminus \{0\}} \Phi'(k) e^{2\pi i k x} \quad (10.6)$$

$$= x + \frac{1}{\pi} \sum_{k=1}^{\infty} \Phi'(k) \sin(2\pi k x) \quad (10.7)$$

$$= x - \frac{1}{\pi} \sum_{m=0}^{\infty} \frac{\Phi(m + \frac{1}{2})}{2m + 1} \sin(2\pi(2m + 1)x). \quad (10.8)$$

Only odd positive integers contribute in the middle series.

Proof. Differentiate (10.2) at $\alpha = 0$ and use (10.3). Since $\Phi(0) = 1$, $\Phi'(0) = 0$, and $\Phi(k) = 0$ for nonzero integers, division by $2\pi i$ gives (10.6). Pair k with $-k$ to obtain the real sine series. If k is even, the zero has order at least two, so $\Phi'(k) = 0$. For odd $k = 2m + 1$, the first sinc factor has a simple zero and

$$\Phi'(2m + 1) = -\frac{\Phi(m + \frac{1}{2})}{2m + 1}, \quad (10.9)$$

which gives (10.8). $\square$

The transform formula (3.12) also yields, for real $\xi \neq 0$,

$$\int_0^1 F(x) \cos(2\pi\xi x) \, dx = \frac{\sin(2\pi\xi) - \Phi(\xi/2) \sin(\pi\xi)}{2\pi\xi}, \quad (10.10)$$

$$\int_0^1 F(x) \sin(2\pi\xi x) \, dx = \frac{\Phi(\xi/2) \cos(\pi\xi) - \cos(2\pi\xi)}{2\pi\xi}. \quad (10.11)$$

At integer frequencies the cosine coefficients vanish, while the sine coefficients are explicit samples of Φ .

Contribution VII: a continuous spectral family

The usual half-integer cosine expansion sits inside a twisted Poisson family parameterized by α . Letting the twist tend to zero converts spectral values into spectral slopes and yields a sine series for F supported exactly on the simple odd zeros of the Rvachev Fourier product.

11 Sparse lobe signs and base- 2^r digit sums

The staircase zero multiplicities of Ψ_r generate a natural automatic sign sequence. Polyphase recombination then factors the ordinary Thue–Morse sign into r such sparse signs.

Let $s_b(n)$ denote the sum of the base- b digits of n .

Theorem 11.1 (Lobe signs of a sparse sinc product). *For $n \geq 0$ and any $x \in (n, n + 1)$,*

$$\operatorname{sgn} \Psi_r(x) = (-1)^{\sum_{j=1}^n m_r(j)} \quad (11.1)$$

$$= (-1)^{n + \sum_{\ell \geq 1} \lfloor n/2^{r\ell} \rfloor} \quad (11.2)$$

$$= \boxed{(-1)^{s_{2^r}(n)}}. \quad (11.3)$$

Thus the signs of the consecutive real lobes of Ψ_r form the parity-of-digit-sum sequence in base 2^r .

Proof. The sign starts positive on $(0, 1)$ and flips once for each zero multiplicity crossed, giving (11.1). Since $m_r(j) = 1 + \lfloor \nu_2(j)/r \rfloor$,

$$\sum_{j=1}^n \left\lfloor \frac{\nu_2(j)}{r} \right\rfloor = \sum_{\ell \geq 1} \#\{j \leq n : 2^{r\ell} \mid j\} = \sum_{\ell \geq 1} \left\lfloor \frac{n}{2^{r\ell}} \right\rfloor.$$

For $b = 2^r$, the base- b Legendre identity is

$$\sum_{\ell \geq 1} \left\lfloor \frac{n}{b^\ell} \right\rfloor = \frac{n - s_b(n)}{b - 1}.$$

Because $b - 1$ is odd and b is even, $n + (n - s_b(n))/(b - 1)$ has the same parity as $s_b(n)$. $\square$

At $r = 1$, (11.3) is the familiar formula $\operatorname{sgn} \Phi(x) = \varepsilon_n$ on the n th lobe. For $r > 1$ it supplies a companion automatic sequence attached to the sparse Rvachev density.

Corollary 11.2 (Polyphase factorization of the Thue–Morse sign). *For every $r \geq 1$ and $n \geq 0$,*

$$\varepsilon_n = \prod_{a=0}^{r-1} (-1)^{s_{2^r}(\lfloor n/2^a \rfloor)}. \quad (11.4)$$

Equivalently,

$$\tau(n) \equiv \sum_{a=0}^{r-1} s_{2^r}(\lfloor n/2^a \rfloor) \pmod{2}. \quad (11.5)$$

The same representation is analytic:

$$\varepsilon_n = \prod_{a=0}^{r-1} \operatorname{sgn} \Psi_r\left(\frac{n + 1/2}{2^a}\right). \quad (11.6)$$

Proof. Take signs in the polyphase product $\Phi(n + 1/2) = \prod_a \Psi_r((n + 1/2)/2^a)$. The argument of the a th factor lies in the lobe indexed by $\lfloor n/2^a \rfloor$. Apply [theorem 11.1](#) and use $\operatorname{sgn} \Phi(n + 1/2) = \varepsilon_n$. The unsigned parity form is the exponent modulo two. $\square$

An entirely arithmetical proof follows by summing the floor exponents:

$$\sum_{a=0}^{r-1} \sum_{m \geq 0} \left\lfloor \frac{n}{2^{a+rm}} \right\rfloor = \sum_{j \geq 0} \left\lfloor \frac{n}{2^j} \right\rfloor = 2n - s_2(n), \quad (11.7)$$

whose parity is $s_2(n)$. Formula (11.4) therefore expresses binary digit parity as a cyclic product of overlapping base- 2^r digit parities.

Corollary 11.3 (Thue–Morse from odd zero slopes). *For every $n \geq 0$,*

$$\varepsilon_n = -\operatorname{sgn} \Phi'(2n + 1), \quad \tau(n) = \frac{1 + \operatorname{sgn} \Phi'(2n + 1)}{2}. \quad (11.8)$$

Proof. Equation (10.9) says $\Phi'(2n + 1) = -\Phi(n + 1/2)/(2n + 1)$. The lobe sign of the half-integer value is ε_n . $\square$

Contribution VIII: a new digital factorization

The sparse component does not merely split amplitudes. Its lobe signs are base- 2^r digit-parity characters, and their r shifted polyphase copies multiply to the binary Thue–Morse character. This gives both a new purely digital formula and a product of analytic sign samples. Odd zero slopes of the full Fourier image give a second pointwise analytic representation.

12 Representation concordance

This section gathers the formulas into a compact lookup index. A reader interested in a particular mode—probability, transform, product, series, operator, geometry, or combinatorics—can use the rightmost column to locate the hypotheses and proof. The list includes both corpus baseline and the new polyphase layer; it is not a novelty list.

12.1 Rvachev’s up-function and its Fourier image

Viewpoint	Representation	Status
Random series	$X = \sum_{n \geq 0} 2^{-n-1} U_n$, with u the density of X .	K
Infinite cube	$u(x) = \int_{[-1,1]^{\mathbb{N}}} \delta(x - \sum 2^{-n-1} v_n) \prod dv_n / 2$ in the distributional sense.	K
Finite convolution	$u = \lim_{N \rightarrow \infty} g_{1,N}$, where $g_{1,N}$ is an explicit nonuniform box spline.	K
Self-CDF	$u(x) = \mathbb{P}(X \leq 2x+1)$ for $-1 \leq x \leq 0$, equivalently $u(x) = F(x+1)$.	K
Delay equation	$u'(x) = 2(u(2x+1) - u(2x-1))$.	K
Derivative cascade	$u^{(N)}(x) = 2^{N(N+1)/2} \sum_{k < 2^N} \varepsilon_k u(2^N x + 2^N - 1 - 2k)$.	K
Fourier inversion	$u(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z x} dz = 2 \int_0^\infty \Phi(z) \cos(2\pi z x) dz$.	K
Characteristic expectation	$\Phi(z) = \mathbb{E} e^{-2\pi i z X}$.	K
Sinc product	$\Phi(z) = \prod_{n \geq 0} \text{sinc}_\pi(z/2^n)$.	K
Triangular cosine product	$\Phi(z) = \prod_{m \geq 1} \cos(\pi z / 2^m)^m$.	K
Canonical zero product	$\Phi(z) = \prod_{n \geq 1} (1 - z^2/n^2)^{1+\nu_2(n)}$.	K
Gamma product	$\Phi(z) = \prod_{n \geq 0} [\Gamma(1 + z/2^n) \Gamma(1 - z/2^n)]^{-1}$.	K
Logarithmic zeta series	$\log \Phi(z) = - \sum_{m \geq 1} \zeta(2m) z^{2m} / [m(1 - 4^{-m})]$, $ z < 1$.	K
Cotangent logarithmic derivative	$\Phi'/\Phi = \sum_{j \geq 0} [(\pi/2^j) \cot(\pi z/2^j) - 1/z]$.	D
Tangent logarithmic derivative	$\Phi'/\Phi = -\pi \sum_{m \geq 1} m 2^{-m} \tan(\pi z/2^m)$.	D
Partial fractions	$\Phi'/\Phi = -2z \sum_{n \geq 1} (1 + \nu_2(n)) / (n^2 - z^2)$.	K
Moment-generating product	$M(s) = \prod_{n \geq 0} \sinh(s/2^{n+1}) / (s/2^{n+1})$.	K
Moment series	$M(s) = \sum_{m \geq 0} \mu_m s^m / m!$, with $\mu_m = \mathcal{B}_m(\kappa_1, \dots, \kappa_m)$.	K
Cumulants	$\kappa_{2m} = B_{2m} / [2m(1 - 4^{-m})]$, $\kappa_{2m+1} = 0$.	K
Critical cosine series	$u(x) = \frac{1}{2} + \sum_{k \geq 0} \Phi(k + \frac{1}{2}) \cos((2k+1)\pi x)$.	K
Partition of unity	$\sum_{j \in \mathbb{Z}} u(x-j) = 1$.	K
Twisted Zak transform	$\sum_n u(x+n) e^{-2\pi i n \alpha} = \sum_k \Phi(k + \alpha) e^{2\pi i (k+\alpha)x}$.	D
Legendre expansion	$u(x) = \sum_{n \geq 0} a_{2n} P_{2n}(x)$, $a_{2n} = \frac{4n+1}{2} \int_{-1}^1 u(x) P_{2n}(x) dx$; all coefficients reduce to moments.	K
Haar, Walsh, and Schauder	Exact rational multiresolution expansions and finite Walsh products are developed in [14].	K
Cauchy transform	$R(z) = \int u(x)/(z-x) dx = \sum_{m \geq 0} \mu_m z^{-m-1}$.	K
Jacobi fraction	$R(z) = 1/(z - a_0 - b_1/(z - a_1 - b_2/(\dots)))$, with coefficients determined by the moment Hankel determinants.	K
Fredholm determinant	$\Phi(z) = \det(I - z^2 T)$ for eigenvalues n^{-2} of multiplicity $1 + \nu_2(n)$.	K

Viewpoint	Representation	Status
Spectral zeta	$\sum_{n \geq 1} (1 + \nu_2(n)) n^{-s} = \zeta(s)/(1 - 2^{-s})$.	K
Heat trace	$K(t) = \sum_{n \geq 1} (1 + \nu_2(n)) e^{-tn^2}$ and $\int_0^\infty t^{s/2-1} K(t) dt = \Gamma(s/2) \zeta(s)/(1 - 2^{-s})$.	D
Polyphase product	$\Phi(z) = \prod_{a=0}^{r-1} \Psi_r(z/2^a)$.	N
Polyphase convolution	$u = *_{a=0}^{r-1} (2^a g_r(2^a \cdot))$.	N
Radon section	$u(t) = \ w_r\ ^{-1} \int_{w_r \cdot y=t} \prod_a g_r(y_a) d\sigma$.	N
Averaging product	$u = (\prod_{j \geq 0} \mathcal{U}_{2^{-j-1}}) \delta_0$.	N
Differential exponential	$u = \exp(\sum_{m \geq 1} B_{2m} D^{2m} / [2m(2m)!(1 - 4^{-m})]) \delta_0$, interpreted as in section 9 .	N

12.2 The bounded and inverse Fabius functions

Viewpoint	Representation	Status
CDF	$F(x) = \mathbb{P}(X \leq 2x - 1)$.	K
Fold	$F(x) = u(x - 1)$ and $F(1 - x) = u(x)$.	K
Symmetry	$F(1 - x) = 1 - F(x)$.	K
Differential equation	$F'(x) = 2F(2x)$ on $0 \leq x \leq 1/2$.	K
Density Fourier integral	$F(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z(x-1)} dz$.	K
CDF sine integral	$F(x) = \frac{1}{2} + \pi^{-1} \int_0^\infty \Phi(z) \sin(2\pi z(2x - 1)) dz/z$.	K
Product integral	Replace Φ in either preceding integral by its sinc, cosine, Gamma, canonical, or polyphase product.	D
Laplace–Stieltjes	$\int_0^1 e^{sx} dF(x) = e^{s/2} M(s/2)$.	K
Ordinary Laplace transform	$\int_0^1 e^{sx} F(x) dx = (e^s - e^{s/2} M(s/2))/s$.	K
Bromwich inversion	$F(x) = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} e^{s(x-1/2)} M(s/2) ds/s$.	D
Fourier image on $[0, 1]$	$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = [e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}]/(2\pi i \xi)$.	D
Cosine transform	Equation (10.10).	D
Sine transform	Equation (10.11).	D
Half-integer cosine series	$F(x) = \frac{1}{2} - \sum_{k \geq 0} \Phi(k + \frac{1}{2}) \cos((2k + 1)\pi x)$.	K
Twisted Poisson family	Equation (10.4), valid for every $\alpha \notin \mathbb{Z}$.	N
Zero-slope sine series	$F(x) = x + \pi^{-1} \sum_{k \geq 1} \Phi'(k) \sin(2\pi k x)$.	N
Half-integer slope series	$F(x) = x - \pi^{-1} \sum_{m \geq 0} \Phi(m + \frac{1}{2}) \sin(2\pi(2m + 1)x)/(2m + 1)$.	N

Viewpoint	Representation	Status
Moment integral	$\int_0^1 x^p F(x) dx = (1 - \mathbb{E}Y^{p+1})/(p+1), \Re p > -1.$	K
Polyphase volume	$F(x) = \int_{w_r \cdot y \leq 2x-1} \prod_a g_r(y_a) dy.$	N
Polyphase section	$F(x) = \ w_r\ ^{-1} \int_{w_r \cdot y = x-1} \prod_a g_r(y_a) d\sigma.$	N
Quantile level set	$G(y) = \int_0^1 \mathbf{1}_{\{F(x) \leq y\}} dx$ for $0 < y < 1.$	D
Quantile transport	$\int_0^1 h(G(y)) dy = \mathbb{E}h(Y).$	K
Quantile moments	$\int_0^1 G(y)^m dy = \mathbb{E}Y^m.$	K
Quantile MGF	$\int_0^1 e^{sG(y)} dy = e^{s/2} M(s/2).$	K
Inverse derivative	$G'(y) = 1/[2u(2G(y) - 1)].$	K
Radon quantile geometry	$\mathcal{C}_{w_r p_r}(2G(y) - 1) = \mathcal{R}_{w_r p_r}(G(y) - 1) = y.$	N
Area duality	$\int_0^1 F(x) dx + \int_0^1 G(y) dy = 1.$	K
Dyadic arithmetic	Finite Gaussian- q -binomial–Thue–Morse formulas for $F(m/2^n)$ are developed in [10, 11].	K
Endpoint Lambert- W	The lower-branch Lambert coordinate and its periodic all-orders expansion are established in the audited asymptotic reports.	K

12.3 Signed and unsigned Thue–Morse representations

Viewpoint	Representation	Status
Binary definition	$\varepsilon_n = (-1)^{s_2(n)}, \tau(n) = (1 - \varepsilon_n)/2.$	K
Recursion	$\varepsilon_{2n} = \varepsilon_n, \varepsilon_{2n+1} = -\varepsilon_n$; equivalently $\tau(2n) = \tau(n), \tau(2n+1) = 1 - \tau(n).$	K
Floor-sum parity	$\varepsilon_n = (-1)^{n - \sum_{j \geq 1} \lfloor n/2^j \rfloor}.$	K
Factorial valuation	$\varepsilon_n = (-1)^{n - \nu_2(n!)}.$	K
Finite coefficient product	$\varepsilon_n = [z^n] \prod_{j=0}^{N-1} (1 - z^{2^j})$ for $n < 2^N.$	K
Block Fourier product	$\sum_{n < 2^N} \varepsilon_n e^{2\pi i n \theta} = \prod_{j < N} (1 - e^{2\pi i 2^j \theta}).$	K
Prouhet moments	$\sum_{n < 2^N} \varepsilon_n n^d = 0$ for $d < N.$	K
Rvachev lobe sign	$\varepsilon_n = \operatorname{sgn} \Phi(n + \frac{1}{2}).$	K
Odd zero slope	$\varepsilon_n = -\operatorname{sgn} \Phi'(2n+1).$	N
Sparse lobe product	$\varepsilon_n = \prod_{a=0}^{r-1} \operatorname{sgn} \Psi_r((n+1/2)/2^a).$	N
Base- 2^r digit factorization	$\varepsilon_n = \prod_{a=0}^{r-1} (-1)^{s_{2^r}(\lfloor n/2^a \rfloor)}.$	N
Unsigned base factorization	$\tau(n) \equiv \sum_{a < r} s_{2^r}(\lfloor n/2^a \rfloor) \pmod{2}.$	N
Spline jump	$\varepsilon_k = q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}.$	N
Unsigned spline jump	$\tau(k) = \frac{1}{2} (1 - q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}).$	N
Distributional comb	$\sum_{k < 2^N} \varepsilon_k \delta_{x_k} = q^{\binom{N}{2}} D^N g_{r,N}.$	N

Viewpoint	Representation	Status
Fourier deconvolution	Equation (7.22).	N
Cardinal integral	$\varepsilon_k = (-1)^N q^{\binom{N}{2}} \int g_{r,N}(x) \varphi_k^{(N)}(x) dx.$	N
Spatial scan	The normalized jumps of $D^{N-1}g_{r,N}$, ordered from left to right, are $\varepsilon_0, \dots, \varepsilon_{2^N-1}.$	N
Signed Laplace product	$\sum_{k < 2^N} \varepsilon_k e^{tx_k} = (-2)^N \prod_{j < N} \sinh(tq^j/2).$	N
Unsigned total product	$\sum_{k < 2^N} e^{tx_k} = 2^N \prod_{j < N} \cosh(tq^j/2).$	N
Evil/odious Fourier products	Equations (7.29) and (7.30).	N
All surviving powers	$\sum \varepsilon_k x_k^{N+m} = (-1)^N q^{\binom{N}{2}} (N+m)! \mu_m^{(r,N)} / m!.$	N
Bell–Bernoulli powers	Substitute (8.9) into the complete Bell polynomial in (8.11).	N
Moment limit	Equation (8.19); $r = 1$ gives all up-moments.	N
Pascal, valuation, and automatic atlas	Many additional binomial, Lucas–Kummer, Catalan, determinant, Mahler, Dirichlet, Walsh, and rarefied-sum formulas are catalogued in [11].	K

How to read the concordance

The atlas is deliberately redundant across mathematical languages. For example, the sinc product, random series, averaging operator, cumulant series, canonical determinant, and Radon section are not six unrelated objects; they are six realizations of one law. The value of the concordance is precisely that a calculation difficult in one realization may become elementary in another.

13 Rigidity and extraction of the polyphase factor

The sparse factor is not an arbitrary choice among many possible entire factors. Once all r phases are required to be dyadic dilates of one normalized entire function, the factor is forced.

Theorem 13.1 (Analytic rigidity of the common polyphase factor). *Let H be entire with $H(0) = 1$ and suppose*

$$\Phi(z) = \prod_{a=0}^{r-1} H(z/2^a) \quad (13.1)$$

for all $z \in \mathbb{C}$. Then

$$H(z) = \Psi_r(z). \quad (13.2)$$

In particular, Ψ_r is the unique normalized entire characteristic function whose first r dyadic dilates convolve to the up-law.

Proof. Near zero, choose the analytic logarithms with value zero. If $\log H(z) = \sum_{n \geq 1} h_n z^n$, then

$$\log \Phi(z) = \sum_{n \geq 1} h_n \left(\sum_{a=0}^{r-1} 2^{-an} \right) z^n.$$

The finite geometric multiplier never vanishes. Hence every h_n is uniquely determined by the Taylor series of $\log \Phi$. Odd coefficients vanish, and for $n = 2m$ one obtains

$$h_{2m} = -\frac{\zeta(2m)}{m(1 - 2^{-2rm})},$$

the coefficients of (4.19). Thus $H = \Psi_r$ near zero; the identity theorem extends equality to the plane. $\square$

There is also an explicit extraction product using only the full Fourier image. Since $\Phi(z)/\Phi(z/2) = \text{sinc}_\pi(z)$,

$$\Psi_r(z) = \prod_{m=0}^{\infty} \frac{\Phi(z/2^{rm})}{\Phi(z/2^{rm+1})}. \quad (13.3)$$

Apparent singularities in individual ratios at common zeros are removable; the ratio is globally the entire sinc factor. Formula (13.3) is a multiplicative scale decimator: it deletes all Fourier scales except those congruent to zero modulo r .

Conversely, the full product is reconstructed by the finite interpolation (4.5). Therefore

$$\Phi \xrightarrow{\text{scale decimation}} \Psi_r \xrightarrow{r \text{ dilates and multiplication}} \Phi \quad (13.4)$$

is an exact analysis–synthesis cycle, analogous to a polyphase filter bank but in the multiplicative algebra of entire characteristic functions.

Contribution IX: uniqueness

The modular-scale component is uniquely determined by the requirement that one entire factor and its first $r - 1$ dyadic dilates multiply to the Rvachev Fourier image. It can be extracted from the full image by an infinite product of adjacent-scale ratios and synthesized back by a finite product.

14 Numerical and exact verification

All identities in the main theorem chain are proved symbolically. The numerical work serves three narrower purposes: detecting normalization errors, checking complex-valued product regrouping at high precision, and making the jump-measure geometry visible. The file `numerical_experiments.py` is included in the archive and uses only Python, mpmath, SymPy, NumPy, and Matplotlib.

14.1 Verification protocol

The script performs four independent checks.

1. It evaluates $\Phi(z)$ and $\prod_{a=0}^{r-1} \Psi_r(z/2^a)$ at real and genuinely complex points for $r = 2, 3, 4, 5$, using 90 decimal digits.
2. It compares the direct finite atomic transform $\sum_k \varepsilon_k e^{-2\pi i z x_k}$ with $(2\pi i z)^N q^{\binom{N}{2}} \Psi_r(z)/\Psi_r(q^N z)$.
3. It constructs finite moments from the hyperbolic MGF in exact rational arithmetic and independently reconstructs them from the signed high power sum (8.5).
4. It verifies the cumulant recombination (4.29) symbolically for orders 2 through 12 and $r = 2, 3, 4, 5$.

Representative exact rows are

Check family	Number of rows	Maximum reported residual
Polyphase product, 90-digit arithmetic	16	7.37×10^{-91}
Fourier deconvolution quotient	9	3.64×10^{-90}
Moment/power-sum reconstruction	18	exactly 0
Cumulant recombination	24	exactly 0

Table 5: Verification summary. The two numerical residuals are at the level of the selected mpmath precision; the other tables use exact SymPy rationals.

(r, N, m)	$\mu_m^{(r,N)}$ from $M_{r,N}$	from signed powers	difference
(1, 6, 2)	455/4096	455/4096	0
(1, 6, 4)	2359721/83886080	same	0
(2, 5, 4)	3161984903/206158430208	same	0
(3, 4, 6)	60145469786596675/24211351596743786496	same	0

The full values are in `power_sum_moment_checks.csv`.

14.2 Visual factorization

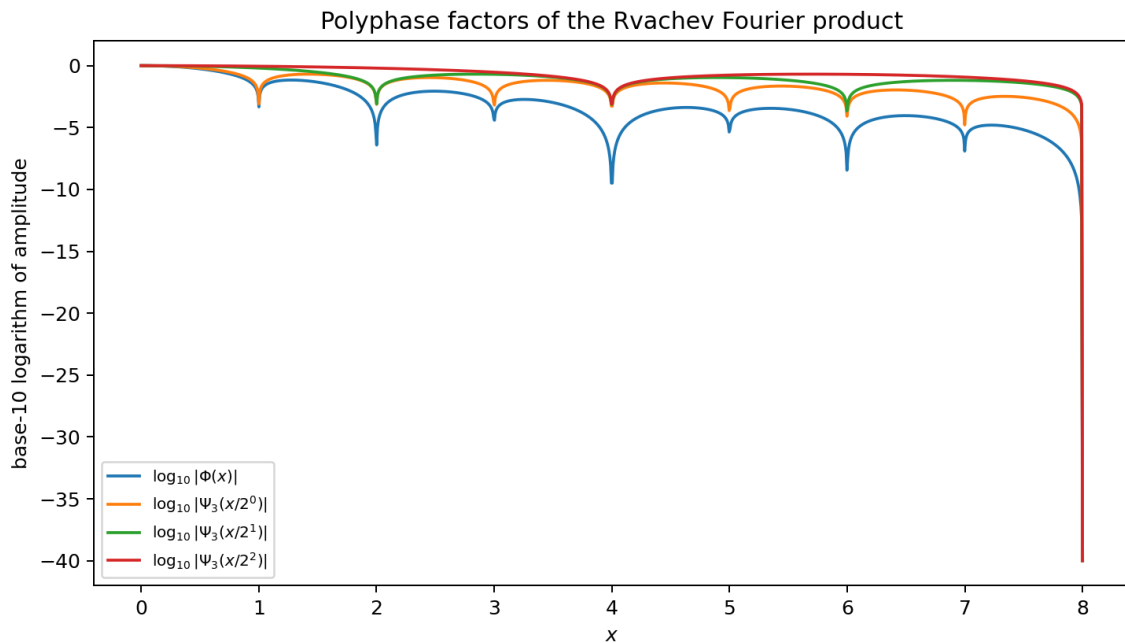


Figure 2: For $r = 3$, the logarithmic magnitude of the full Fourier image and its three sparse factors. At each integer zero the component multiplicities add to the multiplicity of Φ . The graph clips the logarithm near zeros; the product identity itself was checked at high precision away from and near the zero lattice.

The jump-comb visualization in [figure 1](#) is produced from the exact subset sums, not by differentiating noisy sampled data. This matters: the highest classical derivative is piecewise constant and its jumps are mathematically exact Dirac masses in the next distributional derivative.

14.3 Reproduction

From the extracted archive directory, run

```
python numerical_experiments.py
```

The script writes

- polyphase_product_checks.csv;
- fourier_deconvolution_checks.csv;
- power_sum_moment_checks.csv;
- polyphase_cumulant_checks.csv;
- operator_coefficients.csv;
- the two PDF/PNG figures used in this report; and
- verification_log.txt.

No network access, random seed, or machine-dependent numerical integration is required.

Warning 14.1. A tiny residual is not used as a proof of an infinite-product identity. The proofs precede the experiments. The checks are valuable because the formulas contain several easily misplaced factors: 2π , the centered support radius, $q^{\binom{N}{2}}$, the derivative sign $(-1)^N$, and the distinction between $\Psi_r(q^N z)$ and $\Psi_r(z/2^N)$.

15 Conjectures and frontier directions

The statements in this section are separated sharply from the proved results. Some are precise conjectures; others are research programs where fixing the correct normalization is itself part of the work. All concern structures created or clarified by the polyphase decomposition.

15.1 Uniform all-orders deconvolution

Define coefficients $a_{r,\ell}$ by the formal identity

$$\exp\left(-\sum_{m \geq 1} c_{r,m} t^m D^{2m}\right) = \sum_{\ell \geq 0} a_{r,\ell} t^\ell D^{2\ell}. \quad (15.1)$$

Thus

$$a_{r,0} = 1, \quad a_{r,1} = -c_{r,1}, \quad a_{r,2} = -c_{r,2} + \frac{c_{r,1}^2}{2}. \quad (15.2)$$

Conjecture 15.1 (Strong spline-tail expansion). *For fixed $r, m, L \geq 0$,*

$$\left\| D^m g_{r,N} - \sum_{\ell=0}^{L-1} a_{r,\ell} q^{2\ell N} D^{m+2\ell} g_r \right\|_\infty = \mathcal{O}_{r,m,L}(q^{2LN}) \quad (15.3)$$

for all sufficiently large N . In particular,

$$q^{-2N} (g_{r,N}^{(m)} - g_r^{(m)}) \longrightarrow -c_{r,1} g_r^{(m+2)} \quad (15.4)$$

uniformly.

The local Fourier expansion proves the coefficient prediction. A proof of the uniform remainder must use the exact quotient $\Psi_r(z)/\Psi_r(q^N z)$ to control the high-frequency region where the formal reciprocal has poles. Compact support and superalgebraic Fourier decay make the conjecture plausible, but the cancellation near the moving zero lattice should be handled explicitly rather than hidden in a Taylor argument.

15.2 Nondegeneracy of sparse Mellin harmonics

After subtracting the elementary local and mean terms, let P_r denote the periodic fluctuation in the regularized asymptotics of $\log |\Psi_r(t)|$ or $\log M_r(t)$. Normalize its period to r :

$$P_r(\theta) = \sum_{k \in \mathbb{Z}} c_{r,k} e^{2\pi i k \theta / r}. \quad (15.5)$$

Conjecture 15.2 (Mellin-harmonic nondegeneracy). *Apart from coefficients forced to vanish by parity or the chosen regularization, $c_{r,k} \neq 0$ for every $k \neq 0$. Consequently the polyphase projection*

$$P_1(\theta) = r \sum_{\ell \in \mathbb{Z}} c_{r,r\ell} e^{2\pi i \ell \theta} \quad (15.6)$$

loses harmonics only through the exact roots-of-unity cancellation of [theorem 5.1](#), not through accidental zeros of Mellin residues.

A practical attack is to derive a rigorously convergent expression for the regularized Mellin transform $H(s)$ and certify $H(2\pi i k / (r \log 2)) \neq 0$ with interval arithmetic. The decomposition allows one to work at the longer period $r \log 2$, where individual modes may be easier to separate numerically.

15.3 Sparse endpoint asymptotics

The component support radius is $A_r = 1/[2(1 - 2^{-r})]$. General small-deviation heuristics for geometrically weighted sums predict

$$\log g_r(A_r - t) \sim -\frac{(\log(1/t))^2}{2r \log 2}, \quad t \downarrow 0. \quad (15.7)$$

The full Rvachev endpoint theory in the corpus contains much finer lower Lambert– W and periodic structure.

Conjecture 15.3 (Sparse Lambert– W transseries). *For each $r \geq 1$, the endpoint $g_r(A_r - t)$ has an all-orders transseries in a lower-branch Lambert coordinate whose algebraic coefficients are accompanied by a nonconstant periodic function of logarithmic phase with fundamental period $r \log 2$. Under convolution of the r polyphase dilates, the periodic coefficients combine by the Mellin projector of [theorem 5.2](#) to give the ordinary period-log 2 Rvachev expansion.*

This conjecture is stronger than merely replacing $\log 2$ by $r \log 2$ in a leading term. The saddle location, the determinant factor, and the phase origin must all be recomputed for M_r . The reward is an “analysis then synthesis” derivation of the full endpoint fluctuation from sparser components.

15.4 Radon tomography and section algorithms

The equality

$$\mathcal{C}_{w_r p_r}(2x - 1) = \mathcal{R}_{w_r p_r}(x - 1) \quad (15.8)$$

invites several geometric questions.

Open problem 15.4 (Direct section computation). *Develop a certified quadrature for F and G based on the section integral (6.8), compare its conditioning with direct Fourier inversion, and determine the optimal choice of r as a function of the target point and precision.*

For $r = 2$, the integrand is a product of two copies of a broader, flatter component g_2 . Near the endpoints of F , a larger r may distribute the small-deviation event across a higher-dimensional section and improve numerical stability; in the interior, the dimensional cost may dominate. This tradeoff is concrete and testable.

Open problem 15.5 (Quantile tomography). *Use*

$$\mathcal{C}_{w_r} p_r(2G(y) - 1) = \mathcal{R}_{w_r} p_r(G(y) - 1) = y \quad (15.9)$$

to construct monotone Newton or safeguarded secant schemes in which both the residual and derivative are computed as sections of the same product density. Investigate whether the two-level geometry yields a posteriori error bounds stronger than generic inverse-function estimates.

15.5 Orthogonal polynomials and sparse spectral data

Let $(P_n^{(r)})$ be the monic orthogonal polynomials for $g_r(x) dx$. Symmetry gives

$$P_{n+1}^{(r)}(x) = xP_n^{(r)}(x) - \beta_n^{(r)} P_{n-1}^{(r)}(x). \quad (15.10)$$

The moments are explicit Bell–Bernoulli expressions, so every finite $\beta_n^{(r)}$ is an exact rational number.

Open problem 15.6 (Endpoint-flat Jacobi asymptotics). *Determine the asymptotic expansion of $\beta_n^{(r)}$ beyond the universal Nevai limit $A_r^2/4$. In particular, test whether the infinitely flat, log-periodic endpoint behavior of g_r creates logarithmic or periodic corrections to the usual algebraic recurrence-coefficient asymptotics.*

The polyphase convolution gives a second route: the Jacobi data of u should be constrained by the r component moment sequences and their scaled free cumulant-like convolution algebra. No simple formula is expected, but exact Hankel determinants at moderate order may reveal a stable pattern.

15.6 Twisted scale characters

The ordinary polyphase split uses the trivial character of the cyclic group $\mathbb{Z}/r\mathbb{Z}$. Near the origin, where logarithms are single-valued, introduce

$$L_{r,\ell}(z) = \sum_{a=0}^{r-1} \omega^{-a\ell} \log \Psi_r(z/2^a), \quad \omega = e^{2\pi i/r}. \quad (15.11)$$

Then $L_{r,0} = \log \Phi$, while the nontrivial $L_{r,\ell}$ isolate scale characters which cancel in the ordinary product.

Open problem 15.7 (Scale-character continuation). *Find natural single-valued analytic or meromorphic objects whose local logarithms are $L_{r,\ell}$, describe their zero divisors, and determine whether they encode the discarded Mellin harmonics directly. A successful construction would diagonalize modular scale refinement in the same way that the finite Fourier transform diagonalizes cyclic translation.*

Complex powers of factors cannot simply be written globally because zeros create monodromy. A divisor-valued or line-bundle formulation may be the right language.

15.7 General bases and residue subsets

For $b > 1$, the formal analogue

$$\Psi_{b,r}(z) = \prod_{m \geq 0} \text{sinc}_\pi(z/b^{rm}) \quad (15.12)$$

has a base- b polyphase decomposition. Much of the generic base- b theory is already present in the corpus, so the frontier is not the definition itself. The new questions are structural:

- classify residue subsets $A \subset \{0, \dots, r-1\}$ for which the partial product is a characteristic function with especially simple compact support or refinement equation;

- determine the lobe-sign automatic sequence when b is not even, where the parity simplification in (11.3) changes;
- replace binary signs by roots of unity and recover generalized Prouhet–Thue–Morse words from distributional derivatives of complex or vector-valued box splines;
- identify the exact Mellin character projector for arbitrary scale residue weights.

15.8 Arithmetic of sparse knot limits

All finite knot locations, cumulants, moments, and power sums are rational for $q = 2^{-r}$. The limiting density values are subtler.

Open problem 15.8 (Sparse-adic values). *Determine the arithmetic nature of $g_r(x)$ and its derivatives at rational points whose denominators are powers of 2^r . Establish either a rationality theorem with denominator bounds, analogous to the dyadic Fabius theory, or a sharp obstruction showing why the self-CDF mechanism at $r = 1$ is exceptional.*

The plateau present for $r \geq 2$ supplies many trivial rational values, but the transition regions carry the real arithmetic content. The nested factorization (6.19) may transfer information between different r .

15.9 Formalization roadmap

The new theorem chain is unusually well suited to formal verification because most proofs reduce to finite reindexing, characteristic functions, and polynomial distribution identities.

A practical Lean sequence is:

1. define Ψ_r as a locally uniformly convergent entire product and prove the partition $n = a + rm$;
2. construct Z_r from the existing weighted-uniform probability infrastructure and identify its characteristic function;
3. prove the finite and infinite convolution factorizations;
4. formalize zero multiplicities using ν_2 and Euclidean division;
5. define $g_{r,N}$ as a finite convolution, then prove the truncated-power formula and $D^N g_{r,N}$ as a finite signed measure;
6. prove the bit-reversal ordering and parity invariance;
7. derive the power-sum identity by distributional integration by parts, or first in the polynomial finite-difference algebra and then transfer;
8. connect finite cumulants to Bell polynomials and Bernoulli numbers;
9. formalize the base- 2^r lobe-sign theorem using the valuation-counting identity;
10. treat Radon/coarea statements last, since they require the heaviest measure-geometric library support.

The finite jump and power-sum layer can be formalized without first developing all of infinite-product complex analysis, making it a good initial target.

Highest-value next experiments

The most decisive near-term computations are: interval certification of the first sparse Mellin harmonics; high-order exact Jacobi coefficients for g_r ; uniform-error measurements for (15.3); and a direct comparison of Fourier versus Radon-section evaluation of F and G . Each experiment tests

a precise structural prediction and can feed directly into a proof attempt.

16 Conclusion

The Fabius–Rvachev system is unusually representation-rich because a single object lies at the intersection of compactly supported smooth analysis, probability, automatic sequences, infinite products, spectral determinants, and digital asymptotics. The audited repository already develops most of the classical and several frontier realizations. The main purpose of this report has been to add a layer that cuts across those existing themes rather than repeat them.

The central operation is Euclidean division of the Fourier scale index. From that elementary reindexing follow:

1. a sparse characteristic function Ψ_r , its compact density g_r , and a unique polyphase convolution factorization of u ;
2. staircase cosine exponents, quotient-valued zero multiplicities, Gamma and canonical products, a sparse spectral zeta, exact determinant splitting, and cumulant recombination;
3. a roots-of-unity projector on Mellin poles and logarithmic-scale harmonics;
4. a product-density Radon representation in which a Fabius value is both a half-space volume and a parallel hyperplane section;
5. finite sparse splines whose highest derivative jumps, distributional derivative, Fourier quotient, and cardinal integrals all recover Thue–Morse signs;
6. an exact left-to-right reading of the ordinary Thue–Morse word from the spatially ordered jumps;
7. Bell–Bernoulli formulas for every surviving signed power sum and for the separate evil and odious power sums;
8. an infinite product of uniform averaging operators with an exact Bernoulli/zeta differential logarithm;
9. a twisted Poisson family and a zero-slope sine series for F ;
10. a factorization of binary Thue–Morse parity into overlapping base- 2^r digit-parity characters, together with recovery from odd zero slopes of Φ .

Several of these statements are short once the correct object is introduced. That is a strength rather than a weakness: the polyphase factor exposes a common mechanism behind formulas which otherwise look unrelated. The next stage should combine the sparse Mellin analysis, endpoint Lambert– W coordinates, and the exact finite jump calculus. Those three layers offer a credible route from computer-assisted experimentation to theorem-level control of periodic asymptotics and approximation errors.

A Corpus audit boundary

The source boundary is the repository commit

19ad531146cda7bc11a206827566db7f5753d5b3.

The branch advanced while the audit was in progress; pinning the hash prevents later reorganizations from silently changing the novelty comparison.

The recursive inventory covered current expositions, the living frontier synthesis, active thematic drafts, frozen archive lineages, and corrected paper sources. The principal exclusion documents were:

Corpus component	Main themes excluded from novelty claims
Primary exposition	Characterization, probability, Fourier product, dyadic values, derivatives, partition of unity, arithmetic.
Thue–Morse atlas	Signed/unsigned formulas, automatic and Mahler forms, binomial and valuation formulas, Prouhet identities, rarefied sums.
Finite-block frontier	Exact tails, confluent cancellation, finite products, extrapolation, spline approximants.
Representation frontier	Resolvents, logarithmic fixed points, Jacobi fractions, determinants, transform families.
Multiresolution report	Haar, Walsh, Faber–Schauder, beta and sinc mixtures, rational finite systems.
Inverse/sampling report	Quantile transport, inverse self-sampling, level-set and transform formulas.
Exponent/ q report	Newton exponent sequences, Gaussian q -binomial structure, Bell and Bernoulli coefficient algebra.
Spectral/arithmetic report	Zero jets, spectral zeta and determinants, half-integer spectra, arithmetic rays.
Integration reports	Integer/fractional transforms, repeated antiderivatives, monomial products, inverse integrals.
Fourier-decay campaign	Sharp decay scales, orbit dependence, log-periodic fluctuations, audit of competing asymptotics.

The label N therefore means “not located in this pinned TeX corpus after targeted full-tree searches.” It is not a claim that no equivalent formula has ever appeared in the worldwide literature. Targeted public-web searches for the exact combinations “polyphase Rvachev”, “scale-residue sinc factorization”, “Thue–Morse box-spline jumps”, and “Fabius hyperplane section” did not reveal a direct antecedent, but a formal priority claim would require a dedicated bibliographic study across several adjacent literatures.

B Archive contents and reproducibility

The ZIP archive accompanying this report contains:

- `fabius_rvachev_polyphase_representations.tex`, the source;
- `fabius_rvachev_polyphase_representations.pdf`, the rendered report;
- `numerical_experiments.py`, with detailed comments and no network dependency;
- five CSV verification tables;
- two figures in both PNG and PDF form;
- `verification_log.txt`;
- `README.md` with build commands; and
- `MANIFEST.sha256` for integrity checking.

A standard build is

```
python numerical_experiments.py
latexmk -pdf -interaction=nonstopmode \
  fabius_rvachev_polyphase_representations.tex
```

The report itself does not require rerunning the experiments if the included figures are retained.

C Selected identities in machine-friendly form

For use in symbolic systems, the main new identities can be summarized without prose as follows. Put $q = 2^{-r}$.

$$\Psi_r(z) = \prod_{m \geq 0} \operatorname{sinc}_\pi(q^m z), \quad (\text{C.1})$$

$$\Phi(z) = \prod_{a=0}^{r-1} \Psi_r(2^{-a} z), \quad (\text{C.2})$$

$$\log \Psi_r(z) = - \sum_{m \geq 1} \frac{\zeta(2m)}{m(1 - q^{2m})} z^{2m}, \quad (\text{C.3})$$

$$\operatorname{ord}_{z=n} \Psi_r = 1 + \left\lfloor \frac{\nu_2(n)}{r} \right\rfloor, \quad (\text{C.4})$$

$$\mathcal{Z}_r(s) = \frac{\zeta(s)}{1 - 2^{-rs}}, \quad (\text{C.5})$$

$$g_{r,N}(x) = \frac{q^{-\binom{N}{2}}}{(N-1)!} \sum_{k < 2^N} \varepsilon_k (x - x_{r,N,k})_+^{N-1}, \quad (\text{C.6})$$

$$D^N g_{r,N} = q^{-\binom{N}{2}} \sum_{k < 2^N} \varepsilon_k \delta_{x_{r,N,k}}, \quad (\text{C.7})$$

$$\varepsilon_k = q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}, \quad (\text{C.8})$$

$$\widehat{g_{r,N}}(z) = \Psi_r(z) / \Psi_r(q^N z), \quad (\text{C.9})$$

$$\sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+m} = (-1)^N q^{\binom{N}{2}} \frac{(N+m)!}{m!} \mu_m^{(r,N)}, \quad (\text{C.10})$$

$$\varepsilon_n = \prod_{a=0}^{r-1} (-1)^{s_{2^r}(\lfloor n/2^a \rfloor)}, \quad (\text{C.11})$$

$$\varepsilon_n = -\operatorname{sgn} \Phi'(2n+1). \quad (\text{C.12})$$

These formulas use the Fourier convention (2.7); a computer algebra implementation using another convention must modify every 2π factor consistently.

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